



Illuminating
ENGINEERING SOCIETY

TECHNICAL MEMORANDUM:
PROJECTING LONG-TERM
CHROMATICITY COORDINATE SHIFT OF
LED PACKAGES, ARRAYS, AND MODULES
AN AMERICAN NATIONAL STANDARD



**TECHNICAL MEMORANDUM:
PROJECTING LONG-TERM CHROMATICITY
COORDINATE SHIFT OF LED PACKAGES,
ARRAYS, AND MODULES
AN AMERICAN NATIONAL STANDARD**

Publication of this document
has been approved by the IES.
Suggestions for revisions should
be directed to the IES.

**Prepared by:
The IES Testing Procedures Committee**



Copyright 2025 by the Illuminating Engineering Society

Approved by the IES Standards Committee November 24, 2024, as a Transaction of the Illuminating Engineering Society.

Approved February 25, 2025, as an American National Standard.

This document incorporates the edits described in the Errata statement issued on August 13, 2020.

All rights reserved. No part of this publication may be reproduced in any form, in any electronic retrieval system or otherwise, without prior written permission of the IES.

Published by the Illuminating Engineering Society, 85 Broad Street, New York, New York 10004.

IES Standards are developed through committee consensus and produced by the IES Office in New York. Careful attention is given to style and accuracy. If any errors are noted in this document, please forward them to the IES Director Standards, at standards@ies.org or the above address, for verification and correction. The IES welcomes and urges feedback and comments.

ISBN: 978-0-87995-538-0

Printed in the United States of America.

DISCLAIMER

IES publications are developed through the consensus standards development process approved by the American National Standards Institute. This process brings together volunteers representing varied viewpoints and interests to achieve consensus on lighting recommendations. While the IES administers the process and establishes policies and procedures to promote fairness in the development of consensus, it makes no guaranty or warranty as to the accuracy or completeness of any information published herein.

The IES disclaims liability for any injury to persons or property or other damages of any nature whatsoever, whether special, indirect, consequential or compensatory, directly or indirectly resulting from the publication, use of, or reliance on this document.

In issuing and making this document available, the IES is not undertaking to render professional or other services for or on behalf of any person or entity. Nor is the IES undertaking to perform any duty owed by any person or entity to someone else. Anyone using this document should rely on his or her own independent judgment or, as appropriate, seek the advice of a competent professional in determining the exercise of reasonable care in any given circumstances.

The IES has no power, nor does it undertake, to police or enforce compliance with the contents of this document. Nor does the IES list, certify, test or inspect products, designs, or installations for compliance with this document. Any certification or statement of compliance with the requirements of this document shall not be attributable to the IES and is solely the responsibility of the certifier or maker of the statement.

AMERICAN NATIONAL STANDARD

Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria have been met by the standards developer.

Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

The use of American National Standards is completely voluntary; their existence does not in any respect preclude anyone, whether that person has approved the standards or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standards.

The American National Standards Institute does not develop standards and will in no circumstances give an interpretation to any American National Standard. Moreover, no person shall have the right or authority to issue an interpretation of an American National Standard in the name of the American National Standards Institute. Requests for interpretations should be addressed to the secretariat or sponsor whose name appears on the title page of this standard.

CAUTION NOTICE: This American National Standard may be revised at any time. The procedures of the American National Standards Institute require that action be taken to reaffirm, revise, or withdraw this standard no later than five years from the date of approval. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute.

Prepared by the IES Testing Procedures Committee

KC Fletcher, *Chair*

Andrew Jackson, *Vice Chair*

David N. Randolph, *Secretary*

Members

Carl K. Andersen	Po-Chieh Hung	Steven Longo	Jacki S. Swiernik
Randall P. Bergin	Jianzhong Jiao	Lorne Loudin	Jonathan C. Vollers
Rolf S. Bergman	Mihaly Kotrebai	Greg McKee	Yuin Zong
Eric Bretschneider	Becky Kuebler	Saori Mitsuhashi	
Aaron A. Feldman	James E. Leland	Michael Piscitelli	
Michael L. Grather	Kurt M. Liepmann	Emil Radkov	
Jeff N. Hulett	Jeff Lockner	Mark B. Sapcoe	

Advisory Members

Lawrence M. Ayers	Victor Eberhard	C. Cameron Miller	Tobias Schneider
Alex Baker	Phil Elizondo	Yoshihiro Ohno	Anthony W. Serres
Carl A. Bloomfield	James Grandusky	Erik Page	Pratibha Sharma
Fred Carpenter	Kyle J. Hemmi	Doosung Park	Ryan W. Siddon
Pei-Ting Chou	Robert E. Higley	Eric S. Perkins	Daniel Spicer
Troy E. Cowan	Gareth John	Michael Piscitelli	Gary A. Steinberg
Patxi Cruz	Jozsef Juhasz	David P. Ramer	Matthew Stevenson
Lynn Davis	Ryan Kelley	Timothy W. Rasinski	Lane Swainston
Jeff J. Demirjian	Jessica D Kramer	Daniel Rogers	Anders Thorseth
Francisco De Caso	Kelly C. Lerbs	Michael P. Royer	Jason R. Tuenge
Mark E. Duffy	Philip McCarthy	Marvin Ruffin	
Elijah W. Dunn	Greg McKee	Markus Schneider	

CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction	1
1.2	Scope	1
2.0	Normative References	1
2.1	ANSI/IES LM-80-15	1
2.2	ANSI/NEMA/ANSI C78.377-2017	1
2.3	ANSI/IES TM-21-19	2
3.0	Definitions	2
3.1	chromaticity shift, $\Delta u'v'$	2
3.2	chromaticity shift life, CS_n	2
3.3	differential chromaticity, $\delta u'^*, \delta v'^*$	2
3.4	DUT	2
3.5	LED light source	2
3.6	relative chromaticity $\delta u', \delta v'$	2
4.0	Test Data and Sample Size	2
4.1	Data to Be Used	2
4.2	Measurement Interval and Sample Size Recommendations	3
4.3	Chromaticity Data Collection	3
5.0	Chromaticity Coordinate Shift Projection	3
5.1	Method	3
5.2	Procedures	4
5.2.1	Relative Chromaticities	4
5.2.2	Average Relative Chromaticities	4
5.2.3	Differential Chromaticities	4
5.2.4	Data Used for Curve-Fit	4
5.2.5	Curve-Fit	4
5.2.6	Projecting Future Behaviors	5
5.2.7	Adjustment of Results	5
5.2.8	Notation for Projected Chromaticity Shift Life	6

6.0 Temperature and Current Interpolation..... 6

7.0 Report 7

Annex A – Comments on Color Maintenance Behavior 8

Annex B – Types of Chromaticity Shift Behavior 11

Annex C – Mathematical Model Development..... 13

Annex D – Calculation Examples 21

Annex E – Extrapolation Methods36

Additional Reading46

1.0 Introduction and Scope

1.1 Introduction

The Illuminating Engineering Society (IES) has defined standard test methods to be used by the manufacturers of LED products to ensure that the products meet the expectations of the lighting community. One of the most important of these is *ANSI/IES LM-80-21, Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules*. LM-80 provides a test and measurement protocol for how these LED light sources in order to document changes in their light and chromaticity characteristics with respect to time. By applying the methodology of *ANSI/IES TM-21-21, Technical Memorandum: Projecting Long Term Lumen Maintenance of LED Light Sources* to a minimum of 6,000 hours of LM-80-compliant test results, it is possible to project the lumen maintenance life (e.g., L_{70} , L_{80} , or L_{90}) of LED packages, arrays, and modules.

In addition to luminous flux, LM-80 also addresses another key photometric parameter, the chromaticity coordinates (often called the “color point”) of the LEDs. The initial chromaticity coordinate requirements for white solid-state lighting (SSL) light sources and luminaires are defined in ANSI/NEMA/ANSLG C78.377, Specifications for the Chromaticity of Solid State Lighting Products. Proper selection of LEDs used to build SSL products allows luminaire manufacturers to easily achieve a desired ANSI C78.377-defined chromaticity.

It is important to note that in some applications—such as retail and museum lighting—a characterization of anticipated chromaticity shift of SSL products over their usable lifetime is very important to the user. Further, the chromaticity coordinates of some light sources can change significantly from their initial values over time. This document recommends a method of projecting the shift in chromaticity coordinates of LED packages, arrays, and modules using data obtained during LM-80 testing.

1.2 Scope

This document provides recommendations for projecting long-term chromaticity coordinate stability of LED light sources using data obtained per *ANSI/IES LM-80-21, Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays, and Modules*.

This method shall only be used for phosphor converted LEDs with initial chromaticity coordinates that fall within the ranges defined for solid state lighting products in ANSI/NEMA/ANSLG C78.377-2017.

This method shall not be used to predict chromaticity shifts in excess of $\Delta u'v' = 0.010$.

2.0 Normative References

2.1 ANSI/IES LM-80-21

Illuminating Engineering Society. Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules. New York: IES; 2021.

2.2 ANSI/NEMA/ANSLG C78.377-2017

National Electrical Manufacturers Association. Specifications for the Chromaticity of Solid-State Lighting Products. Arlington, Virginia: NEMA; 2017.

2.3 ANSI/IES TM-21-21

Illuminating Engineering Society. Projecting Long Term Lumen Maintenance of LED Light Sources. New York: IES; 2021.

3.0 Definitions

3.1 chromaticity shift, $\Delta u'v'$

The magnitude of change in chromaticity coordinates relative to the initial chromaticity coordinates.

$$\Delta u'v'(t) = \sqrt{[u'(t) - u'(0)]^2 + [v'(t) - v'(0)]^2} = \sqrt{\delta u'(t)^2 + \delta v'(t)^2}$$

3.2 chromaticity shift life, CSn

The elapsed operating time until the LED light source exhibits a chromaticity shift of $\Delta u'v' = n \times 10^{-3}$ from its initial chromaticity coordinates. Examples:

CS4: Time (hours) to $\Delta u'v' = 0.004$

CS7: Time (hours) to $\Delta u'v' = 0.007$

3.3 differential chromaticity, $\delta u'^*$, $\delta v'^*$

The time rate of change of a chromaticity coordinate, per hour; calculated by dividing the change in a chromaticity coordinate over a time interval by the number of hours in the time interval

3.4 DUT

Device under test; the LED light source defined in **Section 3.5**.

3.5 LED light source

LED package, array, or module that is operated via an auxiliary driver. (See **Section 2.1**.)

3.6 relative chromaticity $\delta u'$, $\delta v'$

The change in a chromaticity coordinate relative to the initial value.

4.0 Test Data and Sample Size

4.1 Data to Be Used

The data to be used in this projection method shall be collected according to the methods described in IES LM-80-21. LEDs with significantly different chromaticity coordinates may exhibit different chromaticity shift behaviors; therefore, only data collected for LEDs from a single ANSI C78.377-2017 chromaticity bin shall be used for this method.

Since the changes in DUT chromaticity coordinates between time intervals is typically very small (i.e., $\Delta u'v' < 0.001$), this method shall only be used with LM-80 datasets in which the DUT chromaticity data are reported to at least 5 decimal places. When the average change in relative chromaticity coordinates between time intervals is much less than 0.00005, the predicted trend for chromaticity shift may exceed the actual trend and therefore could be inaccurate. All calculations shall be made using at least 5 significant digits.

4.2 Measurement Interval and Sample Size Recommendations

All data from the sample set at a given case temperature and drive current from the ANSI/IES LM-80-21 test report for a specific product model shall be used for projection of chromaticity shift. A minimum test duration of 7,000 hours is required. The preferred measurement interval for this method is 600 hours or less. The measurement interval shall not exceed 1,000 hours. It should be noted that because the measurement interval determines the number of data points that will be used to predict future behavior, a change in the number of data points will lead to a change in the uncertainty and time interval for projection of chromaticity shift. Accordingly, different extrapolation limits are allowed based on the measurement interval.

The preferred sample size is a minimum of 30 units. As with the measurement interval, a change in sample size leads to a change in uncertainty and therefore results in a different extrapolation limit for projection of chromaticity shift. Specifically:

- For a sample size of 30 units or greater and a measurement interval of 600 hours or less, the extrapolation limit as specified in **Section 5.2.7** for chromaticity shift shall be 6.0 times the total ANSI/IES LM-80-15 test duration. For a sample size of 30 units or greater and a measurement interval greater than 600 hours up to a maximum of 1,000 hours, the extrapolation limit as specified in **Section 5.2.7** for chromaticity shift shall be 5.5 times the total ANSI/IES LM-80-15 test duration.
- For a sample size of 20 units to 29 units and a measurement interval of 600 hours or less, the extrapolation limit as specified in **Section 5.2.7** for chromaticity shift shall be 5.5 times the total ANSI/IES LM-80-15 test duration. For a sample size of 20 units to 29 units and a measurement interval greater than 600 hours up to a maximum of 1,000 hours, the extrapolation limit as specified in **Section 5.2.7** for chromaticity shift shall be 5.0 times the total ANSI/IES LM-80-15 test duration.
- For a sample size of 10 units to 19 units and a measurement interval of 600 hours or less, the extrapolation limit as specified in **Section 5.2.7** for chromaticity shift shall be 5.0 times the total ANSI/IES LM-80-15 test duration. For a sample size of 10 units to 19 units and a measurement interval greater than 600 hours up to a maximum of 1,000 hours, the extrapolation limit as specified in **Section 5.2.7** for chromaticity shift shall be 4.5 times the total ANSI/IES LM-80-15 test duration.

This method for chromaticity shift projection shall not be used for a sample size of fewer than 10 units or a measurement interval greater than 1,000 hours.

4.3 Chromaticity Data Collection

Additional measurements beyond 7,000 hours are encouraged and will provide the basis for more accurate chromaticity shift projection.

The DUT samples tested per ANSI/IES LM-80-15 shall be selected to be sufficiently representative of the overall population of the product. All data collected for the DUT samples tested per ANSI/IES LM-80-15 shall be used for projection, with the exception of failed DUT samples as defined in ANSI/IES LM-80-15. Specially built samples to be used as DUTs shall not be allowed. Selectively excluding data based on performance shall not be allowed.

5.0 Chromaticity Coordinate Shift Projection

5.1 Method

The recommended method of chromaticity shift projection is to apply a curve-fit to the collected data for differential chromaticities, and then to use this information to extrapolate the changes in relative chromaticity to the time point

where the chromaticity shift exceeds an acceptable level (for example CS7 or $\Delta u'v' = 0.007$). That time point is the *chromaticity shift life*. The same curve-fit information for the differential chromaticities can also be used to determine the relative chromaticities, and hence chromaticity shift, at future points in time (e.g., 20,000 hours, 25,000 hours).

Since the differential chromaticity data is calculated using averaged data from the DUTs, data for each DUT shall be complete (i.e., obtained and recorded). Since it is also unlikely that each device under test (DUT) will have the same initial chromaticity coordinates, a change in the number of data points used in calculating the average may result in a significant change in the trend. Therefore, if no chromaticity data are reported for a DUT at a single time interval, then no data for that DUT shall be used for this method. Similarly, if a DUT fails during LM-80 testing, then no data for that DUT shall be used for this method.

This method is applied separately for each set of DUT test data collected at each operational (e.g., drive current) and environmental (e.g., case temperature) condition as specified in LM-80-15.

5.2 Procedures

5.2.1 Relative Chromaticities. The relative chromaticity coordinate data of all samples within the same dataset defined in **Section 5.1** are calculated:

$$\delta u'i(t) = u'i(t) - u'i(0)$$

$$\delta v'i(t) = v'i(t) - v'i(0)$$

5.2.2 Average Relative Chromaticities. The calculated relative chromaticity coordinates within the same dataset defined in **Section 5.1** are averaged for each test condition and measurement interval.

5.2.3 Differential Chromaticities. The differential chromaticities are calculated using the averaged relative chromaticity coordinates within the same dataset as defined in **Section 5.1** for each test condition and each measurement interval:

$$\delta u'^*(t) = [u'(t) - u'(t - 1)]/\Delta t = [\delta u'(t) - \delta u'(t - 1)]/\Delta t$$

$$\delta v'^*(t) = [v'(t) - v'(t - 1)]/\Delta t = [\delta v'(t) - \delta v'(t - 1)]/\Delta t$$

$$\Delta t = t_i - t_{i-1}$$

5.2.4 Data Used for Curve-Fit. All differential chromaticity coordinate data from 2,000 hours forward shall be used for curve-fits.

5.2.5 Curve-Fit. A linear least-squares curve-fit shall be performed for the differential chromaticities as a function of time using (see **Annex D** for calculation examples):

$$\delta u'^*(t) = a_u t + b_u,$$

$$\delta v'^*(t) = a_v t + b_v,$$

where:

t = operating time, hours

$\delta u'^*(t), \delta v'^*(t)$ = differential chromaticity at time t

a_u, a_v = slope derived by the least-squares fit

b_u, b_v = intercept derived by the least-squares fit

5.2.6 Projecting Future Behaviors. Relative chromaticity shift shall be projected using a numerical method, as described here.

Numerical Projection:

Using the information at a reference time and relative chromaticity coordinates, it is possible to calculate the relative chromaticity coordinates as a function of time. The reference time and coordinates shall be those for the last data point used in the calculations (i.e., $\delta u'$ and $\delta v'$ at the final measurement interval). Since the differential chromaticities give the rate of change of the relative chromaticity coordinates as a function of time, they may be used to calculate the change in relative chromaticity coordinates for a given time interval using:

$$\delta u'(t + \Delta t) = \delta u'(t) + \delta u'^*(t)\Delta t = \delta u'(t) + (a_u t + b_u)\Delta t ,$$

and:

$$\delta v'(t + \Delta t) = \delta v'(t) + \delta v'^*(t)\Delta t = \delta v'(t) + (a_v t + b_v)\Delta t ,$$

where Δt shall be the average measurement interval of the dataset used for the calculations.

When performing numerical projections, the time interval used shall be the average of the ANSI/IES LM-80-15 data measurement interval for that condition.

To determine CS_n , the above formulas shall be used in an iterative manner until the chromaticity shift threshold is exceeded. The calculated time shall be obtained using linear interpolation between the two nearest extrapolated time intervals.

Whenever a CS_n value is exceeded experimentally in the course of ANSI/IES LM-80-15 testing, the reported value shall be obtained by linear interpolation between the two nearest test points. The result takes precedence over any value projected by the formulas above.

5.2.7 Adjustment of Results. For a sample size of 30 units or more and a measurement interval of 600 hours or fewer, chromaticity shift shall not be projected beyond 6.0 times the total test duration of the measured data. For a sample size of 20 to 29 units and a measurement interval of 600 hours or fewer, chromaticity shift shall not be projected beyond 5.5 times the total test duration of the measured data. For a sample size of 10 to 19 units and a measurement interval of 600 hours or fewer, chromaticity shift shall not be projected beyond 5.0 time the total test duration of the measured data.

For a sample size of 30 units or more and a measurement interval of more than 600 hours and fewer than or equal to 1,000 hours, chromaticity shift shall not be projected beyond 5.5 times the total test duration of the measured data. For a sample size of 20 to 29 units and a measurement interval of more than 600 hours and fewer than or equal to 1,000 hours, chromaticity shift shall not be projected beyond 5.0 times the total test duration of the measured data. For a sample size of 10 to 19 units and a measurement interval of more than 600 hours and fewer than or equal to 1,000 hours, chromaticity shift shall not be projected beyond 4.5 times the total test duration of the measured data. For a sample size of less than 10 units or a measurement interval of greater than 1,000 hours chromaticity shift shall not be projected.

Table 5-1 summarizes these conditions.

Table 5-1. Allowable Extrapolation Limits of Chromaticity Shift as a Function of Number of Test Devices and Measurement Interval

Test Units	Measurement Interval	
	>600 to ≤1,000 hours	≤600 hours
≥30	5.5	6.0
20 – 29	5.0	5.5
10 – 19	4.5	5.0

Data collected within 48 hours of each measurement point are acceptable for use in product chromaticity shift projections. The exact time value(s) shall be used in the calculations for differential chromaticity parameters. The average measurement time interval shall be used for extrapolation calculations.

5.2.8 Notation for Projected Chromaticity Shift Life. The chromaticity shift life projected in this method shall be expressed using $CSn(Dk)$, where n is 1,000 x chromaticity shift measured using $u'v'$ coordinates, and D is the total duration of the test in hours divided by 1,000 and rounded to the nearest 0.5. Examples:

CS7(7k) for 7,000 hours of test data

CS7(8.5k) for 8,500 hours of test data

If the calculated CSn value is reduced by the extrapolation limits described in **Section 5.2.7**, the color shift life value shall be expressed with a symbol ">". For example:

CS7(8k) > 32,000 hours (at $T_s = 85^\circ\text{C}$, $I_f = 700\text{ mA}$)

If the CSn value is reached experimentally in the course of LM-80-15 testing, the chromaticity shift life shall be expressed with the D value equal to the CSn value in hours and rounded to the nearest 100. For example:

CS7(7k) = 6,200 hours (at $T_s = 105^\circ\text{C}$, $I_f = 700\text{ mA}$)

When a CSn value is reached experimentally in the course of LM-80-15 testing or in extrapolated results, the reported value shall be obtained by linear interpolation between the two nearest data points.

6.0 Temperature and Current Interpolation

No temperature or current interpolation shall be used for projecting chromaticity stability. CSn values shall be calculated using test conditions described in ANSI/IES LM-80-15 that equal or exceed the case temperature and drive current of the LED.

7.0 Report

The report of chromaticity shift life projection shall include the information shown in **Table 7-1** and **Figure 7-1**. The calculated and reported CS4 and CS7 values shall be rounded to the nearest 100 hours. Differential chromaticity coefficients ($a_{u'}$, $b_{u'}$, $a_{v'}$, $b_{v'}$) shall be rounded to 4 significant digits.

A vector graphic showing the relative chromaticity shift lifetimes for CS4 and CS7 shall also be included in the report. Vector graphic information for chromaticity shift lifetimes that exceed the allowable extrapolation limits of **Section 5.2.7** shall not be plotted.

Table 7-1. Recommended Information to Be Included in Each Report at Each ANSI/IES LM-80-15 Test Condition

Description of LED Light Source Tested (manufacturer, model, catalog number)	Value
Nominal ANSI C78.377 Chromaticity bin, K	
Sample size	
Number of failures	
DUT drive current, mA	
Tested case temperature, °C	
Test duration, hours	
Measurement interval, hours	
Reference time, hours	
Reference $\delta u'$	
Reference $\delta v'$	
$\delta u'^*$ slope	
$\delta u'^*$ intercept	
$\delta v'^*$ slope	
$\delta v'^*$ intercept	
Reported CS4, hours	
Reported CS7, hours	

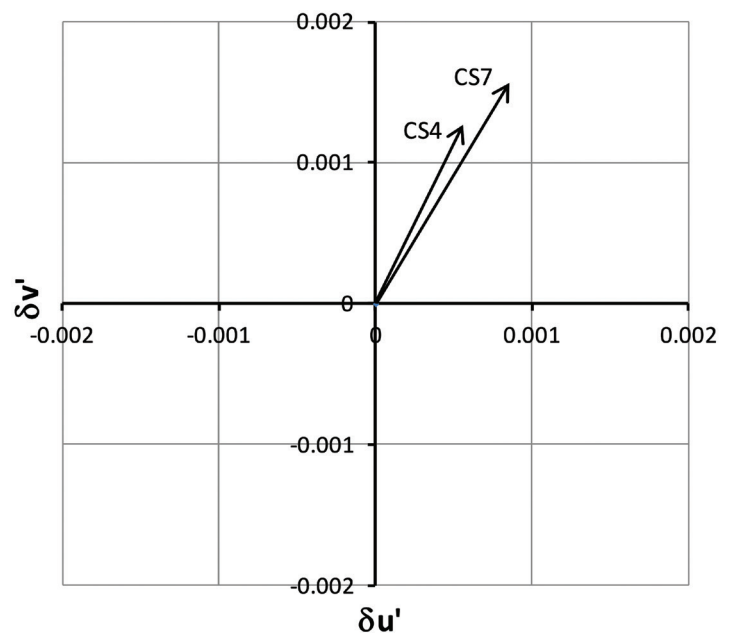


Figure 7-1. Vector graphic showing predicted direction and magnitude of color shift at CS4 and CS7.

Annex A – Comments On Color Maintenance Behavior

Previous to the development of this Technical Memorandum, chromaticity shift specifications for LEDs have been based on not exceeding a specified value over a certain test interval. This decision was based on the practical issues associated with a lack of a predictive model. However, numerous examples exist illustrating that an LED with essentially constant chromaticity coordinate shift over a period of several thousand hours suddenly begins to shift in a consistent (linearly increasing) manner and soon exhibits unacceptable degrees of change in chromaticity coordinates.

This is shown by the LED data in **Figure A-1**, in which $\Delta u'v'$ is approximately equal to 0.0005 from about 1,000 hours until about 6,000 hours, after which it increases linearly at a rate of about 0.0007 per thousand hours. The obvious trend can be projected to reach $\Delta u'v' = 0.007$ at about 15,500 hours.

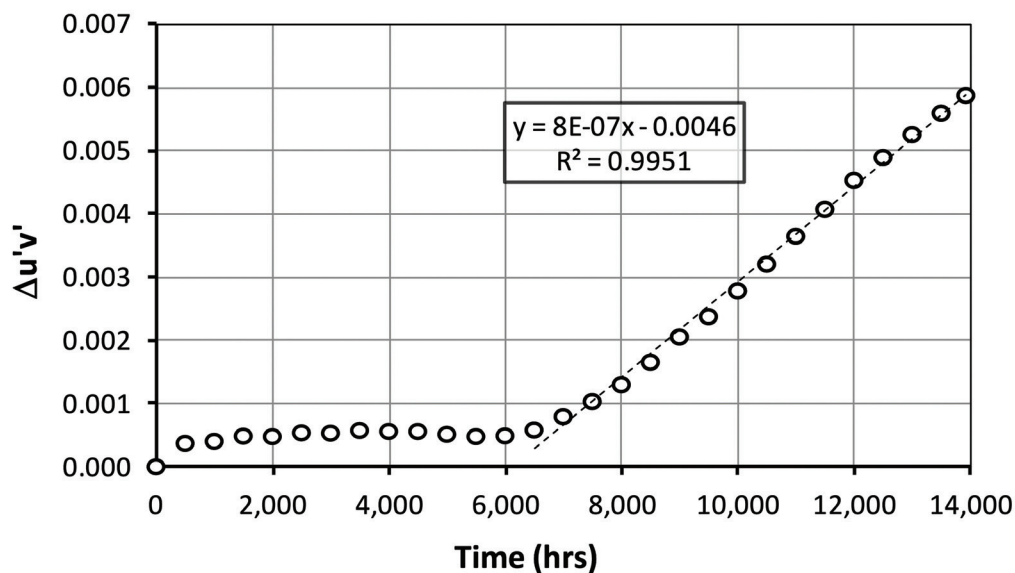


Figure A-1. An example of LED color shift behavior, illustrating an essentially linear increase in $\Delta u'v'$ after a period of about 6,000 hours.

More important, it is possible for at least some LEDs to exhibit extreme shifts in chromaticity coordinates while exhibiting excellent lumen maintenance. Plotting the data from the LED shown in **Figure A-1** against the lumen maintenance data reveals the potential severity of the issue (see **Figure A-2**).

The challenge of developing a model for predicting the chromaticity coordinate shift of LEDs is related to the complicated range of behaviors exhibited by LED products. Chromaticity shift behavior is usually shown as a plot of $\Delta u'v'$ versus time, where $\Delta u'v'$ is calculated as the difference in chromaticity coordinates between a given time and time zero.

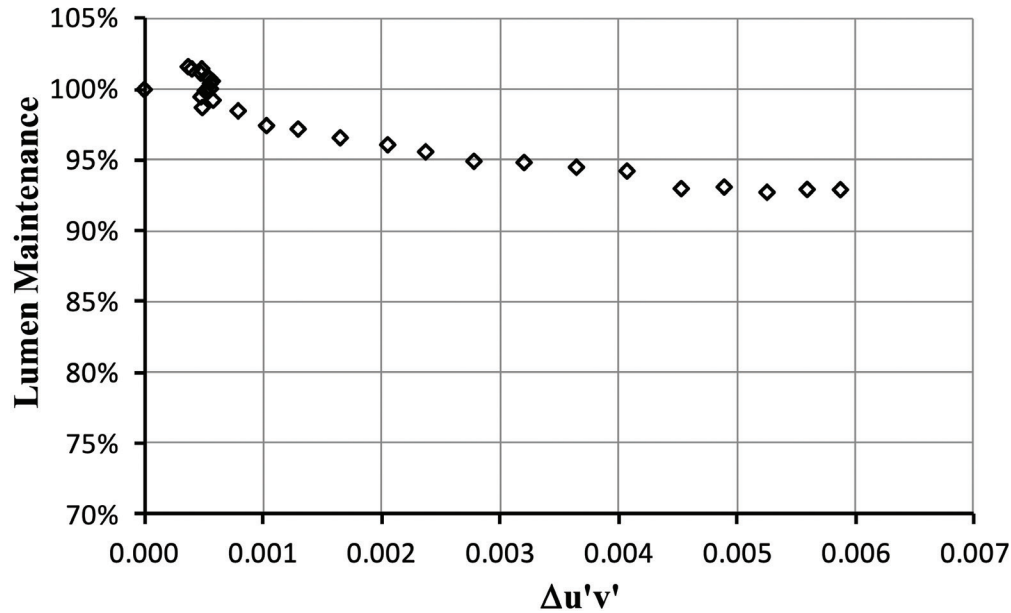


Figure A-2. Lumen maintenance and chromaticity shift behavior of the LED from Figure A1. This LED is projected to reach a chromaticity coordinate shift of $\Delta u'v' = 0.007$ while maintaining lumen maintenance greater than 90%.

A.1 Incubation, Recovery, and Emergence

There are three phases of behavior of LEDs with respect to viewing $\Delta u'v'$ data as a function of time: incubation, recovery, and emergence (see **Figure A-3**). Most LEDs show a rapid short-term increase in $\Delta u'v'$, which is then followed by an extended period during which $\Delta u'v'$ is essentially constant. This period of constant $\Delta u'v'$ is termed the *incubation period*. This may sometimes be followed by a period during which $\Delta u'v'$ decreases with time. This is termed *recovery*. The final phase is characterized by $\Delta u'v'$ behavior that increases as an approximately linear function of time until a chromaticity shift threshold is reached. This is termed *emergence*.

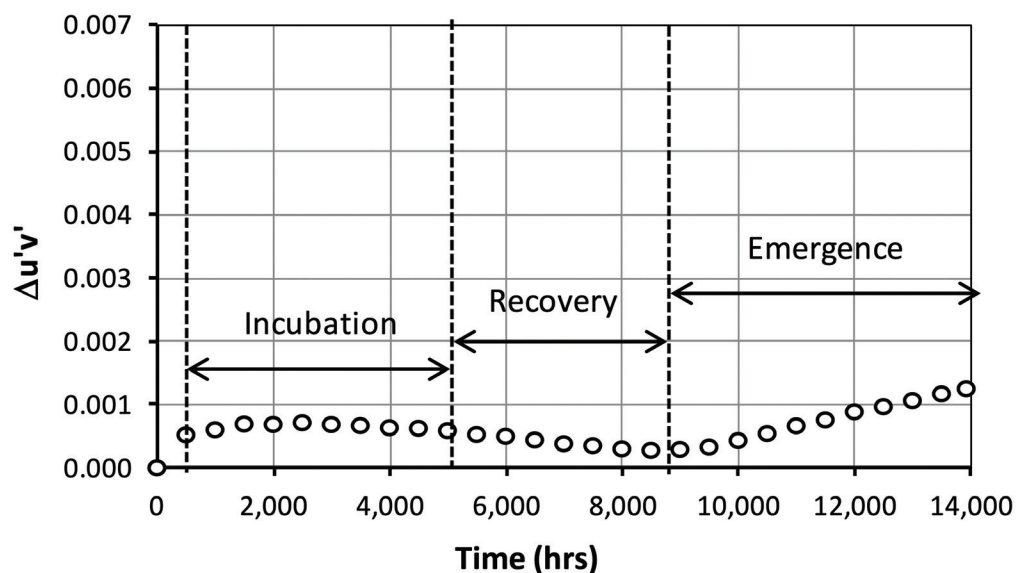


Figure A-3. An example of LED color maintenance behavior illustrating incubation, recovery, and emergence.

LEDs may not exhibit all three phases of behavior. Most LEDs will exhibit incubation, although under extreme operating conditions (e.g., drive current and case temperature near the maximum allowable) some LEDs may reach emergence almost immediately. Other LEDs may exhibit incubation followed by emergence (see **Figure A-4**). However, once an LED enters emergence, the chromaticity shift continues to increase in an approximately linear fashion until it exceeds $\Delta u'v' = 0.007 - 0.010$.

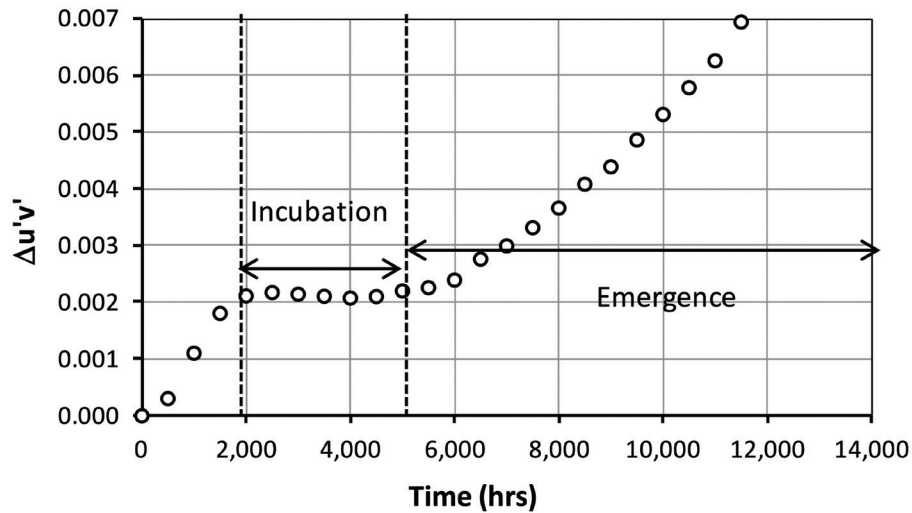


Figure A-4. An example of LED chromaticity coordinate shift behavior showing incubation followed by emergence.

A.2 The Influence of Drive Current and Temperature

Chromaticity shift is a vector (two-dimensional) phenomenon. Reporting a vector quantity using a scalar metric ($\Delta u'v'$) is a form of data compression and may yield a constant scalar quantity associated with a changing vector quantity. During incubation the two dimensional chromaticity shift of an LED may approximate movement along an arc with a constant radius. **Figure A-5** shows the relative chromaticity coordinates of the LED in **Figure A-4**.

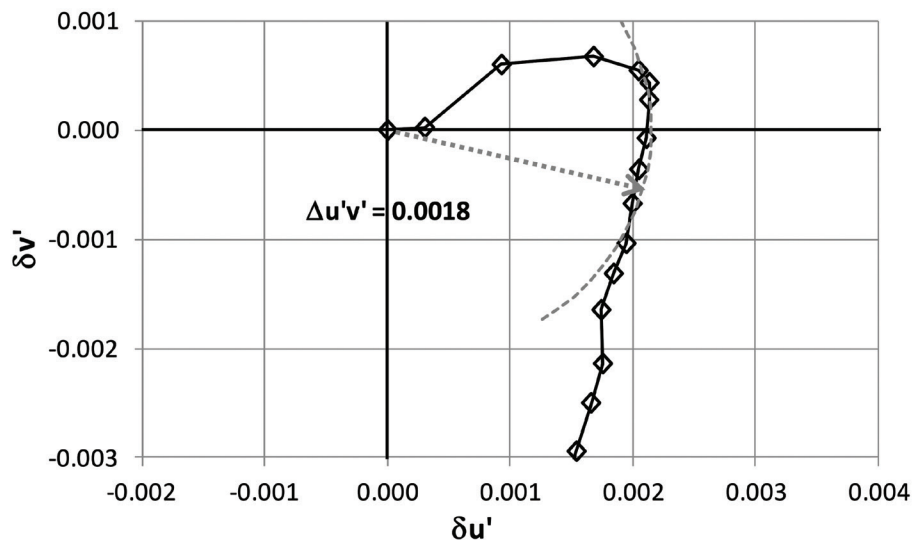


Figure A-5. Chromaticity shift data for the LED of Figure A4. During incubation, the chromaticity coordinates move approximately along a circle of constant radius. When the chromaticity coordinates cease tracking the circular arc, the LED enters the emergence phase.

The time at which emergence begins varies with both drive current and operating temperature. Similarly, the rate at which $\Delta u'v'$ increases during emergence also changes with both drive current and operating temperature. The value of $\Delta u'v'$ during incubation will vary slightly with both drive current and operating temperature. **Figure A-6** provides an example of this behavior.

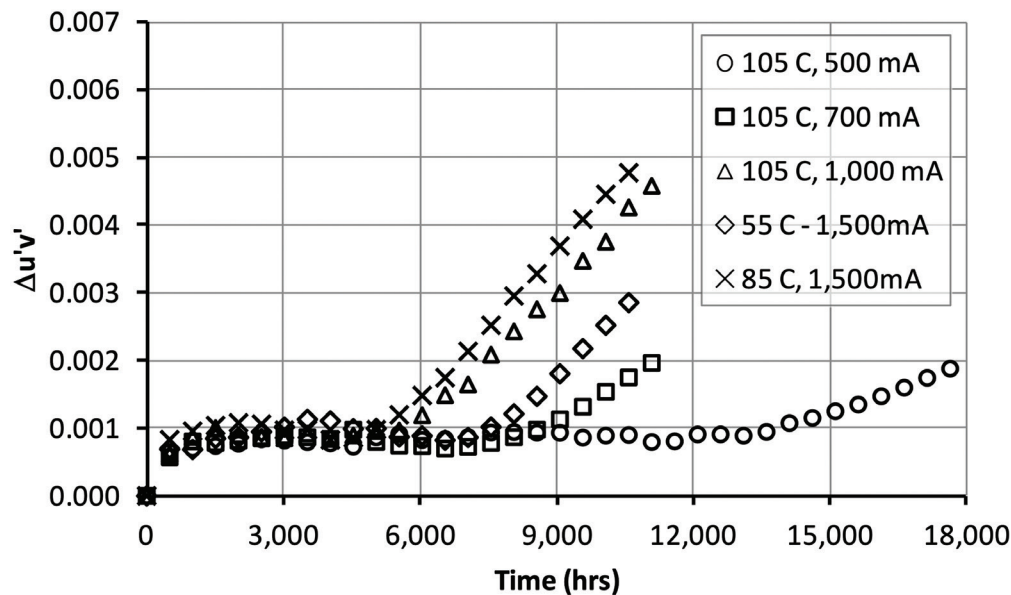


Figure A-6. An example of chromaticity shift behavior for an LED at different operating conditions.

The similarity of measured $\Delta u'v'$ values during incubation is one of the factors that has complicated the development of models capable of predicting long-term chromaticity shift behavior. After emergence, predicting long-term chromaticity shift behavior is relatively straightforward and simple, but as illustrated in **Figure A-6**, this may require testing for periods in excess of 12,000 hours. This is better than testing until end of life, but not significantly so.

The time required for an LED to reach emergence and the rate of increase in $\Delta u'v'$ is related to both the drive current and the operating temperature, but inadequate data was available to develop a reliable interpolation method for either operating temperature or drive current.

Annex B – Types of Chromaticity Shift Behavior

Chromaticity shift in LED devices is a complex process that often involves multiple processes occurring at different operational times during the LED lifetime. Initially, many LEDs tend to show a rapid shift in chromaticity slightly toward either the blue or green direction. This shift typical corresponds to $\Delta u'v'$ of less than 0.003 and occurs within about the first 1,000 hours of operation. The initial chromaticity shift may then be followed by a prolonged chromaticity shift that can be much larger in magnitude. This prolonged, or *terminal*, shift principally occurs in directions that roughly correlate to one of the major colors—red, green, blue, or yellow (see **Figure B-1**)—and can be in a completely different direction from the initial shift. Since the chromaticity coordinates of an LED device are a balance of emissions over the visible spectrum, chromaticity shifts can be caused by either a relative increase or decrease in emissions over

a particular wavelength range. For example, a blue shift can be caused by either an increase in blue emissions from the LED or a relative decrease in yellow emissions from the phosphor. Likewise, a relative increase in green emissions may produce a green shift, while a relative decrease in green emission may produce a red shift.

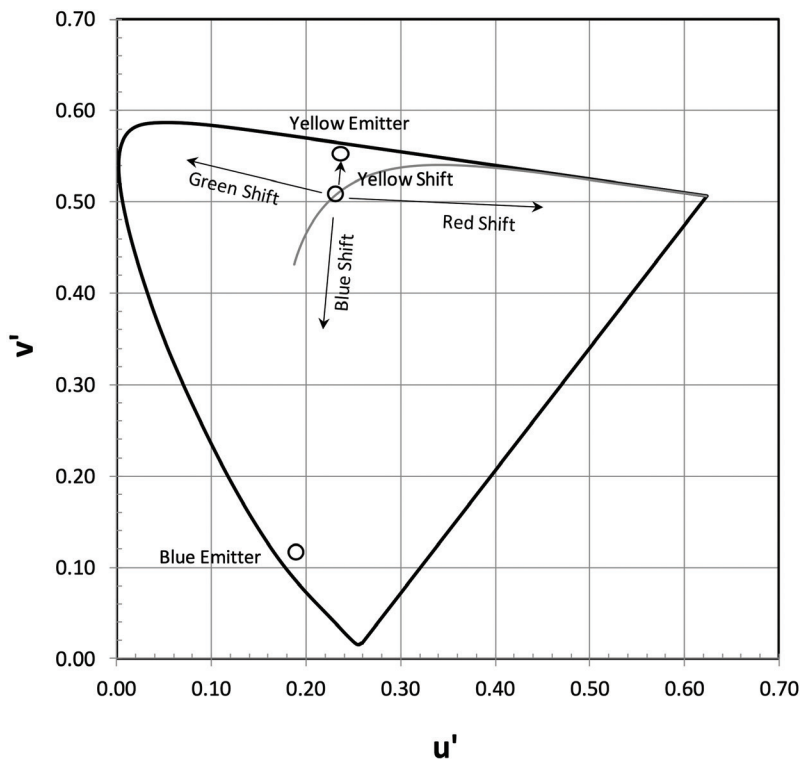


Figure B-1. Chromaticity shift directions for LED devices, shown using the CIE 1976 color space. A representative initial chromaticity point is given by the white circle, and the chromaticity points of the blue and yellow emitters are also shown.

The time required for the impact of the terminal chromaticity shift to be manifested in the device is correlated to the *incubation period* (see **Section A.1**). Since the terminal shift is responsible for the chromaticity shift behavior of the LED device over the majority of its lifetime, understanding this period of prolonged chromaticity shift is especially important in building projections of lifetime chromaticity shift. Developing such projections also requires projecting the length of the incubation time for the terminal chromaticity shift processes.

By examining the chromaticity behavior of LEDs and LED devices, five chromaticity shift modes (CSMs), roughly corresponding to the principal color directions, have been identified in the literature. These CSMs are listed in **Table B-1** and provide guidance on the potential terminal chromaticity shift behaviors that can occur in LED packages.

Table B-1. Chromaticity Shift Modes for LED Devices

Chromaticity Shift Mode	Direction of Terminal Chromaticity Shift	Potential Causes
CSM-1	Blue	Annealing processes in LED package during operation
CSM-2	Green	Oxidation of phosphors
CSM-3	Yellow	Delamination or interior cracks in the binder
CSM-4	Starts in the yellow direction, then turns toward the blue direction	Second (blue) shift is likely caused by photo-oxidation of molding resin
CSM-5	Red	Reduction in quantum efficiency of green phosphors

Annex C – Mathematical Model Development

The TM-35 Working Group (WG) conducted analyses on more than 30 sets of IES LM-80-08 and ANSI/IES LM-80-15 test data, collected from a number of sources. These datasets included multiple drive current and case temperature conditions. Well over 100 individual drive current and case temperature datasets were analyzed. The duration conditions for these datasets ranged from 6,000 hours to 20,000 hours. The LEDs in collective of datasets represented a broad spectrum of package formats and technologies, including PLCC, COB, high-power ceramic, and EMC (sometimes called QFN) packages.

The TM-35 WG investigated a number of different mathematical models. One early proposed model was based on testing LEDs until the trend in emergence had been firmly established. After emergence, a linear fit was used to project $\Delta u'v'$ forward in time. The difficulty with this model was that it required potentially excessive testing, as emergence could take 15,000 hours or more to determine, depending on the LED package construction, materials, drive current, and operating temperature. (See **Figure C-1**.)

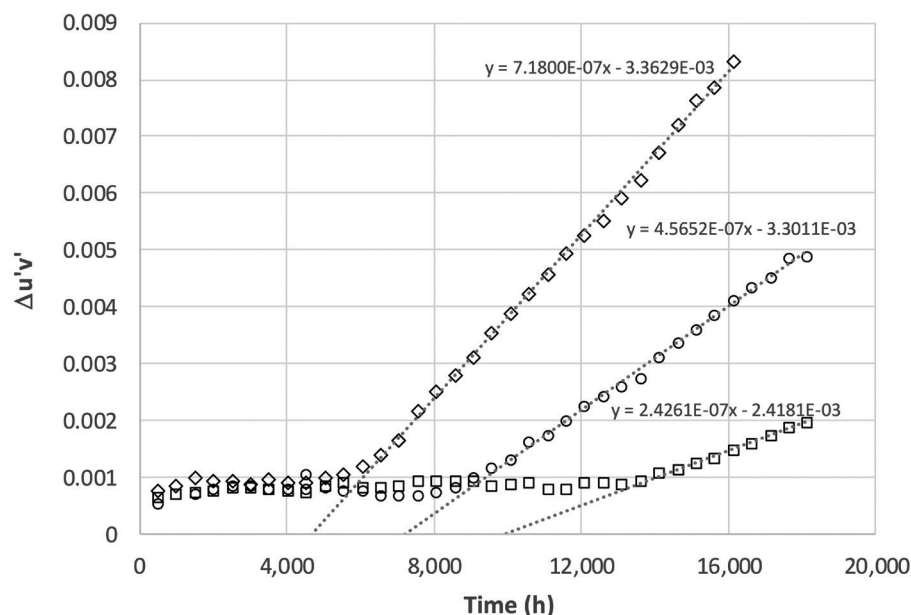


Figure C-1. Linear behavior of $\Delta u'v'$ in emergence phase at different operating conditions. Emergence begins at about 7,000 hours, 9,500 hours and 14,500 hours.

Another model used a step function and assumed a power law behavior for $\Delta u'v'$ after incubation. This model takes the form:

$$\Delta u'v'(t) = \theta(t-d) \cdot a \cdot t^n + c$$

where:

θ is the step function,

d is the incubation period

c is the average value of $\Delta u'v'$ during incubation

a and n are constants that model the behavior of $\Delta u'v'$ after emergence

One complication with the model is that 3 of the parameters (d , a , n) can only be determined after emergence, which equates to a requirement to test until emergence. With enough data points after emergence it is possible this model could predict behavior after emergence to large values of $\Delta u'v'$, but with limited data points the model has only an approximate ability to model behavior.

The model was tested with a dataset as shown in **Figure C-2**. The data from 5,000 to 8,000 hours was used to calculate the parameters by an optimization scheme minimized the difference between the data and the model.

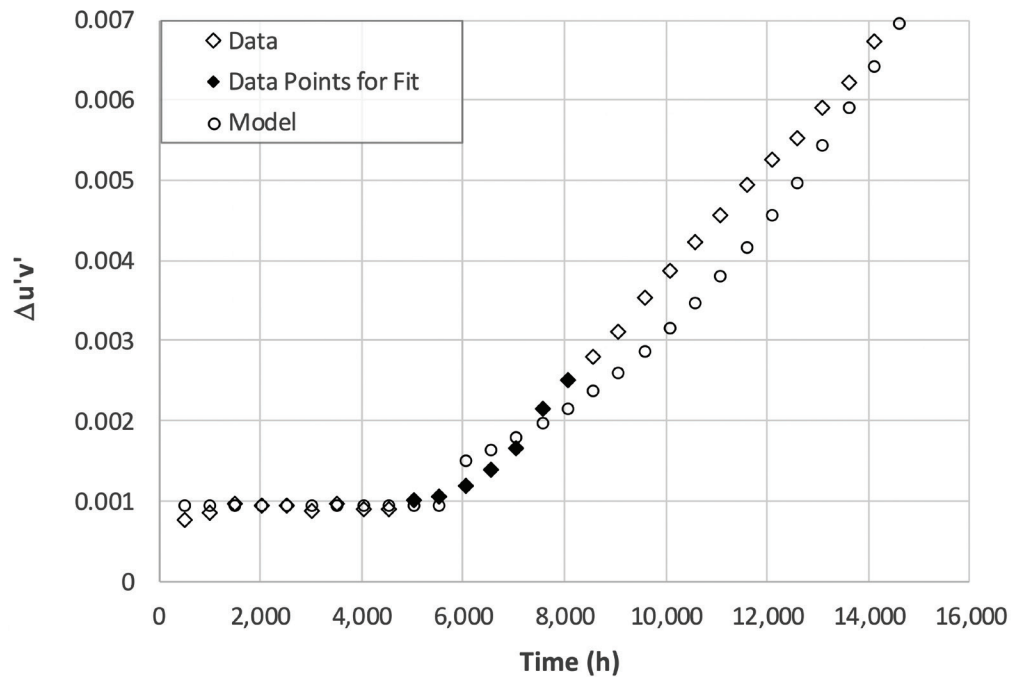


Figure C-2. Evaluation of $\Delta u'v'(t) = \theta(t-d) \cdot a \cdot t^n + c$ model.

After emergence, modeling the chromaticity shift versus time reduces to approximately a linear equation in time. Many models were capable of matching this behavior. Therefore, the ability to predict emergence was chosen as a metric for comparing different models. A benchmark LED dataset was chosen to evaluate several different models. The dataset extended to 12,000 hours and exhibited emergence at just after 6,000 hours. The dataset was chosen based on the proposition that any model incapable of predicting emergence just before it occurred would be unable to accurately predict emergence at some point in the future. A plot of $\Delta u'v'$ versus time for the test dataset is shown in **Figure C-3**.

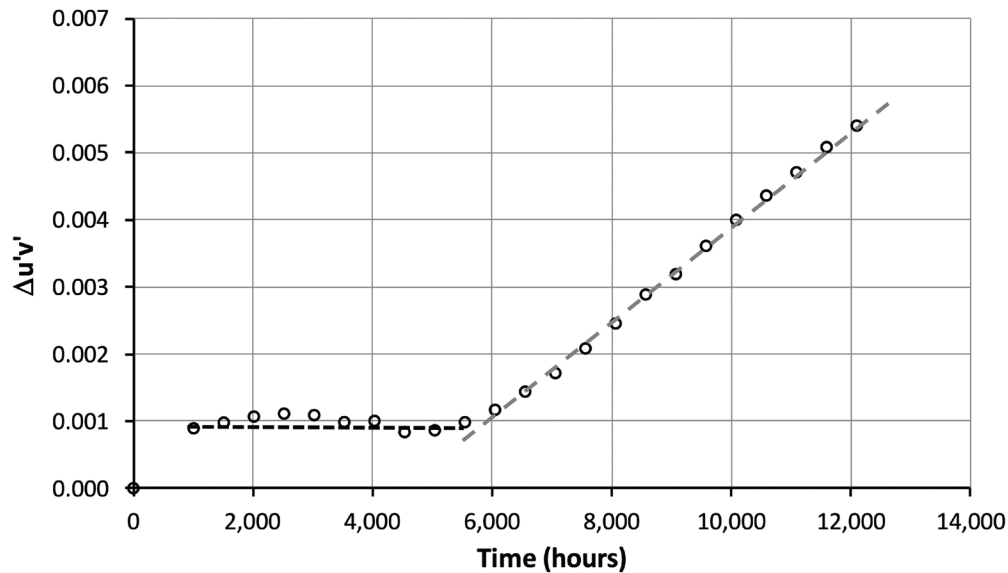


Figure C-3. $\Delta u'v'$ versus time plot of benchmark dataset used to evaluate models. During incubation (short black dashes), $\Delta u'v' = 0.000998 \pm 0.00017$. During emergence (long gray dashes), $\Delta u'v' = 7.2259 \times 10^{-7} \times \text{time} - 0.003315$; $r^2 = 0.9987$.

Most of the models evaluated assumed a linear behavior of chromaticity coordinates with respect to time or a type of exponential decay. This method uses a differential chromaticity analysis (DCA) model that is based on the assumption that differential chromaticity exhibits a linear behavior with respect to time. Some models that were evaluated assumed an exponential behavior of tristimulus values with respect to time. Others required prior knowledge of the chromaticity shift behavior after emergence. Of all the models evaluated, only DCA and derivatives were able to accurately predict the imminent emergence behavior in the test dataset with no assumptions related to the specific package or package format.

In hindsight, chromaticity coordinates may exhibit many different behaviors with respect to time (see **Figure C-4**). They may show a monotonic increase, a monotonic decrease. These increases or decreases may be constant or slowly change rate over time. Other behaviors are possible including an increase followed by a decrease, a decrease followed by an increase. Chromaticity coordinates may also remain essentially constant over long periods of time.

All of these behaviors may be approximated with quadratic functions. The first derivative of a quadratic function is a linear function. Since DCA assumes a linear first derivative, it is mathematically equivalent to a quadratic model for each chromaticity coordinate with respect to time. Since predicting the rate of chromaticity shift is dependent on the rate of change of chromaticity coordinates with respect to time, DCA only calculates the parameters associated with that rate of change.

The differential aspect of the model provides some robustness against noise in the data and is mathematically less complex, as it requires inverting a 2x2 matrix for each chromaticity coordinate as opposed to inverting a 3x3 matrix for each chromaticity coordinate.

Additionally, since only the shift relative to the chromaticity at time = 0 is considered, DCA uses relative chromaticity coordinates – coordinates that are referenced to the values at time = 0. This is a form of normalization and allows averaging of data between different LEDs. Relative chromaticity coordinates also all the behavior of different LEDs to be show on the same plot without loss of detail.

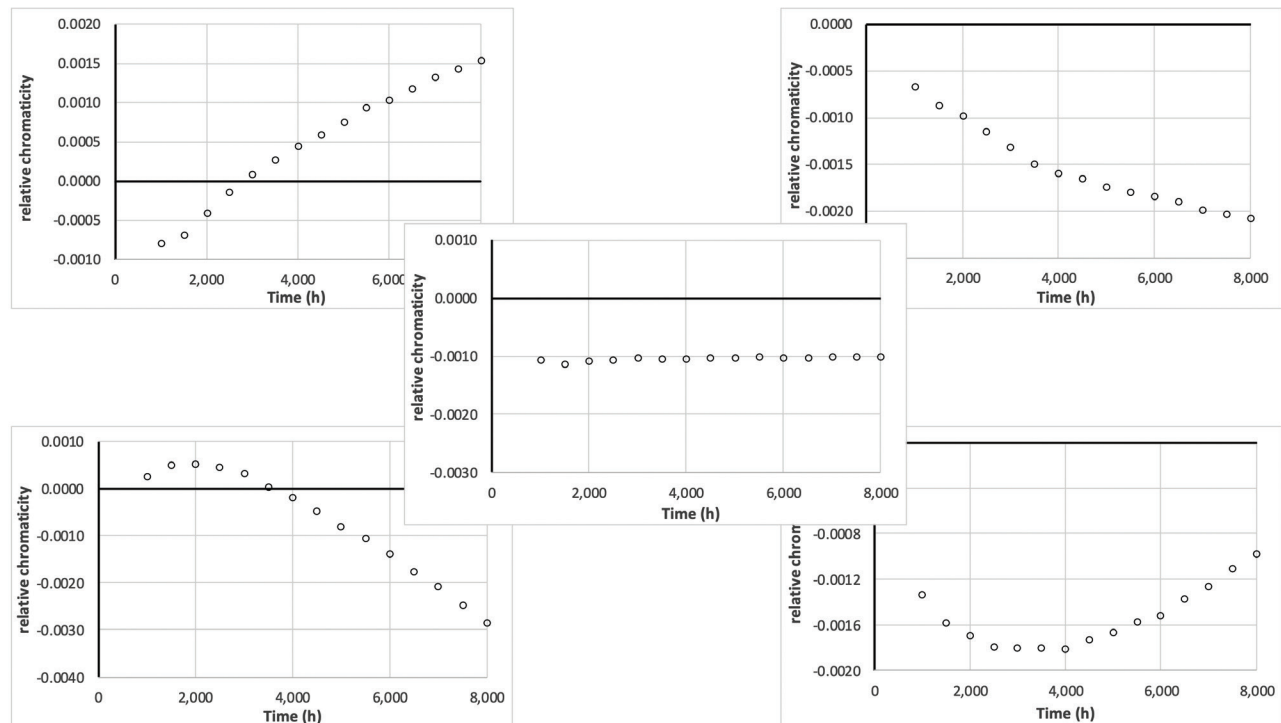


Figure C-4. Examples of range of behavior for relative chromaticity versus time.

By itself, the mathematically simpler approach of differential chromaticity might not be considered sufficient reason to use DCA as a prediction method. Indeed, the WG considered a variation of DCA that yielded an analytical expression for differential chromaticity. This method involved calculating a quadratic fit of the relative chromaticities and then taking the analytical derivative to yield the same linear differential chromaticity parameters (a_u , b_u , a_v , b_v). This approach lacked some appeal, as it required calculation of 6 parameters and then discarded 2 of them.

More important, an in depth review of the datasets revealed that DCA possessed a remarkable robustness against measurement errors that was not present in any of the other models, including the analytical derivative model.

C.1 Selection of Starting Point

Initially, the choice of starting the modeling using the data at 2,000 hours was based on observation. Across all the sets of data analyzed by the WG, it was noted that there were 2 phases of chromaticity shift behavior. There was an initial, rapid shift in chromaticity, which was followed by a long-term slow shift. In examining datasets that included multiple temperatures and drive currents, it was noted that the fast shift behavior was usually completed by about 1,000 hours. The direction and magnitude of this fast shift was noted to be similar, with slight variations correlated to changes in operating temperature and drive current. Since calculating differential chromaticity requires two data points, the first time interval for curve fitting for this method, by default, had to be after the 1,000-hour data point.

Following the reasoning of TM-21-11, which used 6 data points for a linear curve, DCA therefore required a minimum total test duration of 7,000 hours in order to guarantee a minimum of 6 data points for curve fitting.

The second reason for beginning analysis at the 2,000-hour data point came from more-detailed investigations, using long-term datasets that extended past 12,000 hours. Both CS4 and CS7 were projected using data out to 6,000 hours and then progressively longer durations. This was done starting the analysis at the 1,000-hour data point and at the 2,000-hour data point.

For some datasets, there was no significant difference in the projected times for CS4 and CS7, whether or not the 1,000-hour data point was included. In other cases, there was a significant difference in the calculated times for CS4 and CS7. An example case is shown in **Figure C-5**.

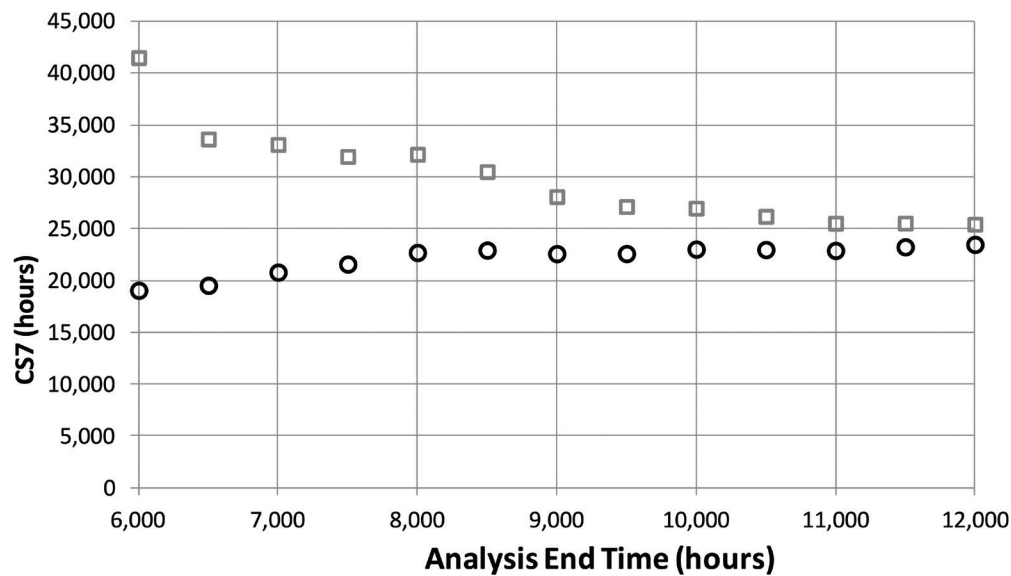


Figure C-5. Comparison of calculated value of CS7, starting analysis at the 1,000-hour (□) and 2,000-hour (○) data points for different durations of analysis.

While both analyses converged at the same value of approximately 24,000 hours, the analysis that began with the 1,000-hour data point started significantly higher and then decreased. In contrast, the analysis that began with the 2,000-hour data point started slightly lower than 24,000 hours and then increased.

This type of trend, of converging estimates that slightly increased when beginning the analysis with the 2,000-hour data point, was noted across the datasets analyzed by the WG. The shorter duration datasets tended to give conservative estimates that improved as the total test duration was increased.

The contrasting behavior between a portion of the 1,000-hour analyses and the 2,000-hour analyses led to the adoption of 2,000 hours as the first data point in DCA calculations.

It is also noted that DCA uses a backwards difference equation when calculating differential chromaticity. This means that for data with a 1,000-hour measurement interval, the initial chromaticity data (i.e., data collected at time = 0 hours) would be used in the calculations. As noted in **Annex B**, LEDs commonly exhibit a chromaticity shift within the first 1,000 hours that is unrelated to long-term chromaticity shift behavior. The magnitude of this shift per unit of time is often much larger than that exhibited at later periods of time and thus may introduce a significant error in the DCA calculations.

C.2 Duration of Data Used

During the development of DCA, it was noted that the method was capable of accurately fitting data over extremely long time spans. An example of a dataset extending past 18,000 hours is shown in **Figure C-6**. This ability to match behavior over long time spans, coupled with the convergent behavior for predicted times to chromaticity shifts discussed above, led the WG to recommend using all data collected at and after 2,000 hours for predictions.

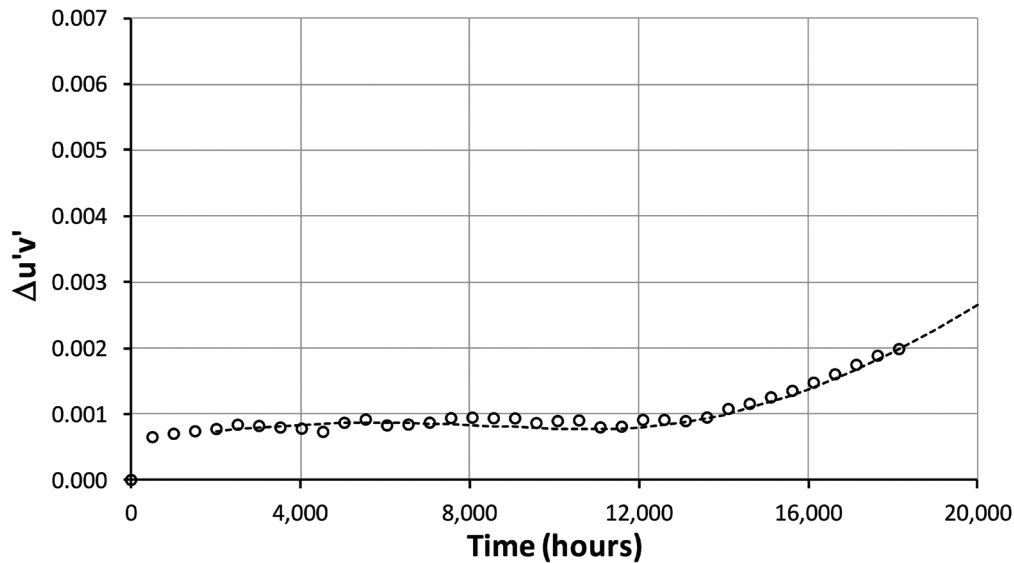


Figure C-6. Plot of $\Delta u'v'$ behavior as a function of time for an 18,114-hour dataset (○). The dashed line represents the result of DCA analysis, using all data gathered starting at 2,000 hours.

The mean variance between the actual $\Delta u'v'$ values and the values extrapolated from the 2,000 data point was 0.0000456 over a 16,000-hour period of time. When the behavior is extrapolated from the 2,000 hours data point, the correlation coefficient over the entire interval is $r^2 > 0.97$.

C.3 Limit for Duration of Prediction

In order to evaluate the accuracy of the DCA method, a model capable of replicating observed chromaticity shift behavior was developed. The model included all of the behaviors exhibited in the datasets. These included an initial fast shift in chromaticity as well as variations in DUT behavior.

The overall trajectory of each DUT in chromaticity coordinates was highly consistent, however differences in the rate at which DUTs travelled along this trajectory were noted to vary. This was evident when the standard deviation of the chromaticity coordinates for a set of DUTs was plotted vs $\Delta u'v'$. The model thus included slight variability in the overall trajectory as well as different speeds along the trajectory for each DUT. The variability was adjusted to match the trends in standard deviation of chromaticity coordinates vs $\Delta u'v'$ seen in the datasets.

Measurement uncertainty was introduced for each DUT using data from the NVLAP Round 15 Proficiency Test. This gave values for the variance for u' and v' were estimated at 0.000200 and 0.000082, respectively. Variances were assumed to be normally distributed.

Monte Carlo simulations were conducted using the model for different behaviors, time intervals and numbers of DUTs. The nominal trajectory for a given run was held constant while the rate of travel along the chromaticity trajectory was changed in order to estimate the accuracy of the model at different extrapolation limits.

Ideal CS7 values were calculated by modeling a set of DUTs without any measurement uncertainty. These idealized values were compared to the calculated values based on a set of DUTs to which an uncertainty was applied for each simulated measurement. By comparing these values, the WG was able to estimate the confidence limit at different extrapolation levels.

Simulations were run using measurement intervals of 500 hours or 1,000 hours and 10 DUTs, 20 DUTs and 30 DUTs, with a total test duration that ranged from 7,000 to 10,000 hours. Idealized CS7 values range from about 12,000 hours to about 84,000 hours, which corresponded to extrapolation limits that ranged between about 1.2 and about 10.5.

Analysis of the dataset corresponding to 20 DUTs, total test duration of 7,000 hours and measurement intervals of 500 hours is shown in **Figure C-7**. As a first comment, it is noted that DCA is unlikely to under-predict the rate of chromaticity shift. This shows the conservative nature of the extrapolation. When the duration of testing is extended past 7,000 hours, the accuracy of the prediction is increased.

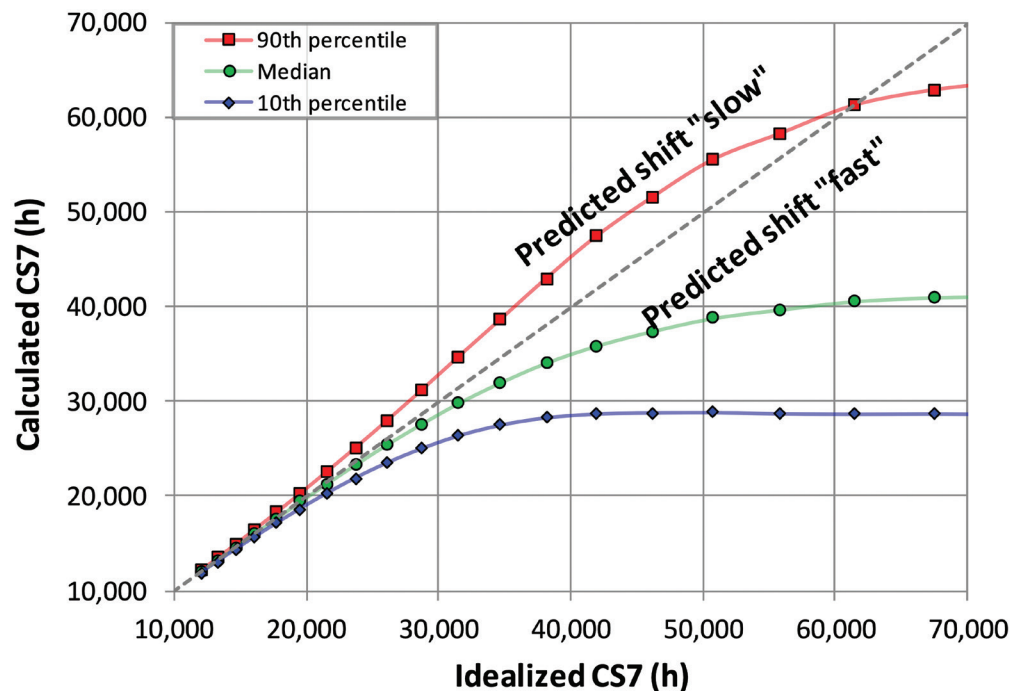


Figure C-7. Error analysis for 20 DUTs, 7,000-hour test duration, 500-hour measurement intervals.

The second feature to be noted is that the confidence bands approach a horizontal asymptote as CS7 increases. These asymptotes are correlated to the measurement uncertainty and represent a region where the prediction of the model is dominated by the measurement uncertainty. This provides a clear indication of one method for determining an extrapolation limit: the intersection of the 90% one-sided confidence limit and the uncertainty boundary. As can be seen in **Figure C-8**, this occurs at an extrapolation limit of about 5.5 for 20 DUTs, 7,000-hour test duration and 500-hour measurement intervals.

In addition, the mean error of the prediction was considered. The geometric mean of the 90% one-sided error estimate was plotted as a function of extrapolation limit. The extrapolation limit for TM-21 was determined based on the 90% one-sided confidence limit. The metric chosen corresponded to an error estimate of about 15%. Applying a mean error limit of 15% was found to yield an extrapolation limit that was virtually identical to the intersection of the 90% one-sided confidence limit and the uncertainty boundary as can be seen in **Figure C-9**. Both of these criteria were used to determine the extrapolation limits of **Section 5.2.7**.

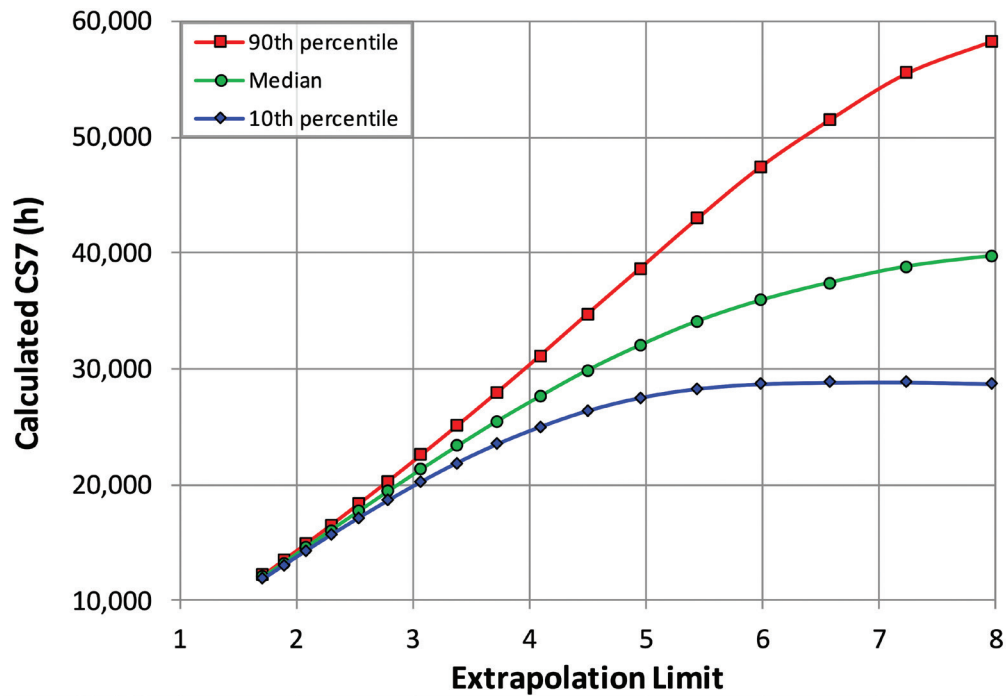


Figure C-8. Error analysis for 20 DUTs, 7,000-hour test duration, 500-hour measurement intervals, plotted as a function of extrapolation limit.

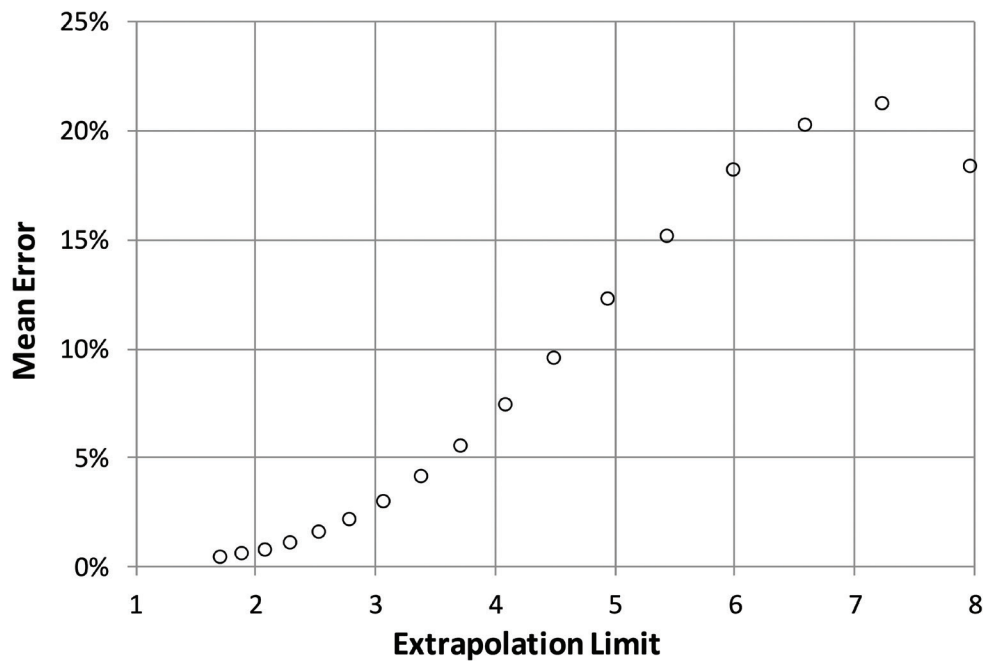


Figure C-9. Mean error estimate versus extrapolation limit for 20 DUTs, 7,000-hour test duration and 500-hour measurement interval.

Annex D – Calculation Examples

To assist users of this document in carrying out calculations following the procedures in **Section 5**, an example set of calculations is presented here. The dataset used is from one of the LM-80 datasets used by the TM-35 working group (WG) in the development of this method. It is recommended that user use the data and calculation results presented here to make a comparison with his or her own calculations to ensure that the calculation steps taken are correct.

For this first example, the operating conditions of the LED were a case temperature of 105 °C and drive current of 500 mA.

Tables D-1a,b and **D-2a,b** provide example u' and v' data, respectively, for the first example LED.

Table D-1a. u' Data for First Example LED.

CIE u'	Sample #									
Time, h	1	2	3	4	5	6	7	8	9	10
0	0.24540	0.24530	0.24722	0.24450	0.24684	0.24859	0.24393	0.24592	0.24781	0.24289
504	0.24489	0.24483	0.24677	0.24395	0.24628	0.24808	0.24341	0.24541	0.24726	0.24228
1,008	0.24487	0.24486	0.24665	0.24393	0.24627	0.24808	0.24339	0.24545	0.24729	0.24229
1,512	0.24482	0.24483	0.24665	0.24392	0.24626	0.24802	0.24332	0.24540	0.24722	0.24222
2,016	0.24477	0.24480	0.24660	0.24382	0.24619	0.24799	0.24327	0.24537	0.24718	0.24228
2,520	0.24472	0.24471	0.24654	0.24376	0.24620	0.24800	0.24323	0.24534	0.24715	0.24212
3,024	0.24476	0.24477	0.24656	0.24377	0.24623	0.24799	0.24324	0.24534	0.24719	0.24219
3,528	0.24477	0.24481	0.24655	0.24383	0.24618	0.24809	0.24335	0.24540	0.24720	0.24222
4,032	0.24475	0.24479	0.24664	0.24383	0.24628	0.24808	0.24332	0.24541	0.24726	0.24225
4,536	0.24484	0.24490	0.24667	0.24385	0.24636	0.24810	0.24338	0.24547	0.24728	0.24232
5,040	0.24468	0.24474	0.24651	0.24372	0.24619	0.24796	0.24320	0.24533	0.24718	0.24213
5,544	0.24468	0.24470	0.24651	0.24375	0.24620	0.24796	0.24320	0.24533	0.24713	0.24213
6,048	0.24475	0.24472	0.24659	0.24375	0.24622	0.24795	0.24319	0.24533	0.24716	0.24218
6,552	0.24474	0.24477	0.24649	0.24379	0.24619	0.24796	0.24322	0.24538	0.24716	0.24214
7,056	0.24473	0.24471	0.24654	0.24371	0.24621	0.24795	0.24321	0.24545	0.24719	0.24213
7,560	0.24469	0.24467	0.24637	0.24363	0.24614	0.24781	0.24318	0.24534	0.24714	0.24204
8,064	0.24462	0.24462	0.24643	0.24354	0.24607	0.24777	0.24313	0.24530	0.24706	0.24204
8,568	0.24466	0.24460	0.24640	0.24355	0.24609	0.24772	0.24308	0.24529	0.24707	0.24207

Table D-1b. u' Data for First Example LED.

CIE u'	Sample #									
Time, h	11	12	13	14	15	16	17	18	19	20
0	0.24590	0.24546	0.24667	0.24641	0.24376	0.24857	0.24686	0.24677	0.24819	0.24677
504	0.24530	0.24488	0.24609	0.24579	0.24313	0.24799	0.24640	0.24617	0.24754	0.24608
1,008	0.24526	0.24483	0.24609	0.24572	0.24311	0.24792	0.24634	0.24618	0.24751	0.24603
1,512	0.24525	0.24482	0.24603	0.24570	0.24310	0.24794	0.24632	0.24614	0.24748	0.24603
2,016	0.24516	0.24479	0.24600	0.24570	0.24302	0.24793	0.24628	0.24625	0.24741	0.24608
2,520	0.24514	0.24472	0.24596	0.24564	0.24304	0.24786	0.24627	0.24612	0.24742	0.24599
3,024	0.24510	0.24476	0.24595	0.24569	0.24307	0.24784	0.24629	0.24616	0.24744	0.24600
3,528	0.24515	0.24480	0.24603	0.24567	0.24309	0.24792	0.24634	0.24614	0.24750	0.24603
4,032	0.24514	0.24485	0.24601	0.24570	0.24309	0.24794	0.24630	0.24617	0.24749	0.24606
4,536	0.24521	0.24485	0.24605	0.24575	0.24320	0.24795	0.24636	0.24623	0.24747	0.24613
5,040	0.24516	0.24481	0.24597	0.24564	0.24312	0.24787	0.24630	0.24611	0.24743	0.24611
5,544	0.24503	0.24471	0.24591	0.24559	0.24303	0.24781	0.24621	0.24608	0.24740	0.24594
6,048	0.24511	0.24474	0.24596	0.24564	0.24309	0.24783	0.24627	0.24612	0.24743	0.24598
6,552	0.24510	0.24474	0.24593	0.24565	0.24307	0.24782	0.24626	0.24613	0.24743	0.24599
7,056	0.24509	0.24467	0.24599	0.24564	0.24312	0.24781	0.24623	0.24620	0.24734	0.24601
7,560	0.24506	0.24463	0.24581	0.24554	0.24296	0.24766	0.24620	0.24621	0.24730	0.24593
8,064	0.24492	0.24456	0.24584	0.24553	0.24302	0.24764	0.24614	0.24612	0.24719	0.24589
8,568	0.24500	0.24456	0.24581	0.24546	0.24295	0.24758	0.24609	0.24613	0.24715	0.24589

Table D-2a. v' Data for First Example LED.

CIE v'	Sample #									
Time, h	1	2	3	4	5	6	7	8	9	10
0	0.52141	0.52133	0.51646	0.51862	0.52183	0.51751	0.52018	0.52315	0.52373	0.52050
504	0.52089	0.52098	0.51602	0.51819	0.52147	0.51729	0.51977	0.52296	0.52351	0.52013
1,008	0.52087	0.52100	0.51600	0.51805	0.52145	0.51720	0.51961	0.52298	0.52343	0.52010
1,512	0.52084	0.52092	0.51592	0.51804	0.52148	0.51712	0.51955	0.52298	0.52344	0.52006
2,016	0.52075	0.52090	0.51590	0.51804	0.52140	0.51715	0.51948	0.52289	0.52344	0.52012
2,520	0.52077	0.52084	0.51577	0.51799	0.52138	0.51704	0.51952	0.52290	0.52337	0.52002
3,024	0.52080	0.52083	0.51589	0.51801	0.52135	0.51711	0.51958	0.52290	0.52336	0.51998
3,528	0.52078	0.52088	0.51584	0.51800	0.52136	0.51709	0.51952	0.52287	0.52336	0.51998
4,032	0.52080	0.52084	0.51580	0.51805	0.52136	0.51702	0.51964	0.52287	0.52330	0.52000
4,536	0.52082	0.52090	0.51584	0.51808	0.52133	0.51712	0.51970	0.52287	0.52335	0.51998
5,040	0.52075	0.52078	0.51580	0.51800	0.52129	0.51699	0.51948	0.52279	0.52322	0.51989
5,544	0.52068	0.52078	0.51576	0.51795	0.52123	0.51703	0.51945	0.52275	0.52324	0.51981
6,048	0.52085	0.52084	0.51591	0.51802	0.52140	0.51707	0.51952	0.52291	0.52331	0.51997
6,552	0.52074	0.52084	0.51588	0.51809	0.52135	0.51713	0.51946	0.52284	0.52333	0.51988
7,056	0.52075	0.52071	0.51594	0.51794	0.52136	0.51708	0.51945	0.52282	0.52327	0.51983
7,560	0.52079	0.52078	0.51587	0.51798	0.52132	0.51700	0.51940	0.52284	0.52327	0.51974
8,064	0.52089	0.52082	0.51597	0.51802	0.52140	0.51709	0.51958	0.52289	0.52333	0.51984
8,568	0.52098	0.52081	0.51608	0.51805	0.52147	0.51713	0.51953	0.52296	0.52332	0.51994

Table D-2b. v' Data for First Example LED.

CIE v'	Sample #									
Time, h	11	12	13	14	15	16	17	18	19	20
0	0.52072	0.52183	0.51997	0.51901	0.51903	0.51614	0.52068	0.52286	0.52579	0.52280
504	0.52022	0.52163	0.51966	0.51855	0.51870	0.51573	0.52048	0.52266	0.52566	0.52253
1,008	0.52024	0.52158	0.51960	0.51843	0.51859	0.51575	0.52037	0.52261	0.52552	0.52248
1,512	0.52020	0.52154	0.51959	0.51842	0.51862	0.51569	0.52033	0.52264	0.52551	0.52244
2,016	0.52015	0.52152	0.51957	0.51838	0.51856	0.51572	0.52028	0.52274	0.52549	0.52253
2,520	0.52009	0.52149	0.51954	0.51843	0.51850	0.51563	0.52034	0.52257	0.52549	0.52243
3,024	0.52008	0.52148	0.51953	0.51836	0.51849	0.51566	0.52034	0.52257	0.52544	0.52239
3,528	0.52009	0.52149	0.51952	0.51838	0.51847	0.51567	0.52030	0.52254	0.52542	0.52242
4,032	0.52012	0.52148	0.51953	0.51844	0.51853	0.51561	0.52031	0.52257	0.52548	0.52244
4,536	0.52012	0.52148	0.51953	0.51845	0.51849	0.51565	0.52031	0.52254	0.52545	0.52244
5,040	0.52005	0.52143	0.51947	0.51835	0.51839	0.51561	0.52026	0.52248	0.52537	0.52242
5,544	0.51998	0.52146	0.51938	0.51838	0.51835	0.51558	0.52033	0.52244	0.52538	0.52231
6,048	0.52012	0.52149	0.51956	0.51848	0.51851	0.51566	0.52036	0.52259	0.52540	0.52242
6,552	0.52006	0.52151	0.51954	0.51856	0.51844	0.51567	0.52034	0.52252	0.52543	0.52234
7,056	0.52004	0.52136	0.51948	0.51851	0.51836	0.51568	0.52035	0.52251	0.52531	0.52228
7,560	0.52010	0.52144	0.51938	0.51854	0.51831	0.51562	0.52035	0.52259	0.52531	0.52232
8,064	0.52010	0.52146	0.51952	0.51855	0.51833	0.51565	0.52045	0.52262	0.52529	0.52226
8,568	0.52012	0.52147	0.51959	0.51867	0.51845	0.51563	0.52045	0.52265	0.52535	0.52229

The first step in the calculations is calculating the relative chromaticity coordinates for the each LED at each time interval. The relative chromaticity coordinates are calculated using:

$$\delta u'(t)_i = u'(t)_i - u'(0)_i$$

$$\delta v'(t)_i = v'(t)_i - v'(0)_i$$

The results are shown in **Tables D-3a,b** and **D-4a,b**.

Table D-3a. $\delta u'$ Relative Chromaticity Coordinates for the First Example LED.

$\delta u'$	Sample #									
Time, h	1	2	3	4	5	6	7	8	9	10
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
504	-0.00051	-0.00047	-0.00045	-0.00055	-0.00056	-0.00051	-0.00052	-0.00051	-0.00055	-0.00061
1,008	-0.00053	-0.00044	-0.00057	-0.00057	-0.00057	-0.00051	-0.00054	-0.00047	-0.00052	-0.00060
1,512	-0.00058	-0.00047	-0.00057	-0.00058	-0.00058	-0.00057	-0.00061	-0.00052	-0.00059	-0.00067
2,016	-0.00063	-0.00050	-0.00062	-0.00068	-0.00065	-0.00060	-0.00066	-0.00055	-0.00063	-0.00061
2,520	-0.00068	-0.00059	-0.00068	-0.00074	-0.00064	-0.00059	-0.00070	-0.00058	-0.00066	-0.00077
3,024	-0.00064	-0.00053	-0.00066	-0.00073	-0.00061	-0.00060	-0.00069	-0.00058	-0.00062	-0.00070
3,528	-0.00063	-0.00049	-0.00067	-0.00067	-0.00066	-0.00050	-0.00058	-0.00052	-0.00061	-0.00067
4,032	-0.00065	-0.00051	-0.00058	-0.00067	-0.00056	-0.00051	-0.00061	-0.00051	-0.00055	-0.00064
4,536	-0.00056	-0.00040	-0.00055	-0.00065	-0.00048	-0.00049	-0.00055	-0.00045	-0.00053	-0.00057
5,040	-0.00072	-0.00056	-0.00071	-0.00078	-0.00065	-0.00063	-0.00073	-0.00059	-0.00063	-0.00076
5,544	-0.00072	-0.00060	-0.00071	-0.00075	-0.00064	-0.00063	-0.00073	-0.00059	-0.00068	-0.00076
6,048	-0.00065	-0.00058	-0.00063	-0.00075	-0.00062	-0.00064	-0.00074	-0.00059	-0.00065	-0.00071
6,552	-0.00066	-0.00053	-0.00073	-0.00071	-0.00065	-0.00063	-0.00071	-0.00054	-0.00065	-0.00075
7,056	-0.00067	-0.00059	-0.00068	-0.00079	-0.00063	-0.00064	-0.00072	-0.00047	-0.00062	-0.00076
7,560	-0.00071	-0.00063	-0.00085	-0.00087	-0.00070	-0.00078	-0.00075	-0.00058	-0.00067	-0.00085
8,064	-0.00078	-0.00068	-0.00079	-0.00096	-0.00077	-0.00082	-0.00080	-0.00062	-0.00075	-0.00085
8,568	-0.00074	-0.00070	-0.00082	-0.00095	-0.00075	-0.00087	-0.00085	-0.00063	-0.00074	-0.00082

Table D-3b. $\delta u'$ Relative Chromaticity Coordinates for the First Example LED.

$\delta u'$	Sample #									
Time, h	11	12	13	14	15	16	17	18	19	20
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
504	-0.00060	-0.00058	-0.00058	-0.00062	-0.00063	-0.00058	-0.00046	-0.00060	-0.00065	-0.00069
1,008	-0.00064	-0.00063	-0.00058	-0.00069	-0.00065	-0.00065	-0.00052	-0.00059	-0.00068	-0.00074
1,512	-0.00065	-0.00064	-0.00064	-0.00071	-0.00066	-0.00063	-0.00054	-0.00063	-0.00071	-0.00074
2,016	-0.00074	-0.00067	-0.00067	-0.00071	-0.00074	-0.00064	-0.00058	-0.00052	-0.00078	-0.00069
2,520	-0.00076	-0.00074	-0.00071	-0.00077	-0.00072	-0.00071	-0.00059	-0.00065	-0.00077	-0.00078
3,024	-0.00080	-0.00070	-0.00072	-0.00072	-0.00069	-0.00073	-0.00057	-0.00061	-0.00075	-0.00077
3,528	-0.00075	-0.00066	-0.00064	-0.00074	-0.00067	-0.00065	-0.00052	-0.00063	-0.00069	-0.00074
4,032	-0.00076	-0.00061	-0.00066	-0.00071	-0.00067	-0.00063	-0.00056	-0.00060	-0.00070	-0.00071
4,536	-0.00069	-0.00061	-0.00062	-0.00066	-0.00056	-0.00062	-0.00050	-0.00054	-0.00072	-0.00064
5,040	-0.00074	-0.00065	-0.00070	-0.00077	-0.00064	-0.00070	-0.00056	-0.00066	-0.00076	-0.00066
5,544	-0.00087	-0.00075	-0.00076	-0.00082	-0.00073	-0.00076	-0.00065	-0.00069	-0.00079	-0.00083
6,048	-0.00079	-0.00072	-0.00071	-0.00077	-0.00067	-0.00074	-0.00059	-0.00065	-0.00076	-0.00079
6,552	-0.00080	-0.00072	-0.00074	-0.00076	-0.00069	-0.00075	-0.00060	-0.00064	-0.00076	-0.00078
7,056	-0.00081	-0.00079	-0.00068	-0.00077	-0.00064	-0.00076	-0.00063	-0.00057	-0.00085	-0.00076
7,560	-0.00084	-0.00083	-0.00086	-0.00087	-0.00080	-0.00091	-0.00066	-0.00056	-0.00089	-0.00084
8,064	-0.00098	-0.00090	-0.00083	-0.00088	-0.00074	-0.00093	-0.00072	-0.00065	-0.00100	-0.00088
8,568	-0.00090	-0.00090	-0.00086	-0.00095	-0.00081	-0.00099	-0.00077	-0.00064	-0.00104	-0.00088

Table D-4a. $\delta u'$ Relative Chromaticity Coordinates for the First Example LED.

$\delta u'$	Sample #									
Time, h	1	2	3	4	5	6	7	8	9	10
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
504	-0.00052	-0.00035	-0.00044	-0.00043	-0.00036	-0.00022	-0.00041	-0.00019	-0.00022	-0.00037
1,008	-0.00054	-0.00033	-0.00046	-0.00057	-0.00038	-0.00031	-0.00057	-0.00017	-0.00030	-0.00040
1,512	-0.00057	-0.00041	-0.00054	-0.00058	-0.00035	-0.00039	-0.00063	-0.00017	-0.00029	-0.00044
2,016	-0.00066	-0.00043	-0.00056	-0.00058	-0.00043	-0.00036	-0.00070	-0.00026	-0.00029	-0.00038
2,520	-0.00064	-0.00049	-0.00069	-0.00063	-0.00045	-0.00047	-0.00066	-0.00025	-0.00036	-0.00048
3,024	-0.00061	-0.00050	-0.00057	-0.00061	-0.00048	-0.00040	-0.00060	-0.00025	-0.00037	-0.00052
3,528	-0.00063	-0.00045	-0.00062	-0.00062	-0.00047	-0.00042	-0.00066	-0.00028	-0.00037	-0.00052
4,032	-0.00061	-0.00049	-0.00066	-0.00057	-0.00047	-0.00049	-0.00054	-0.00028	-0.00043	-0.00050
4,536	-0.00059	-0.00043	-0.00062	-0.00054	-0.00050	-0.00039	-0.00048	-0.00028	-0.00038	-0.00052
5,040	-0.00066	-0.00055	-0.00066	-0.00062	-0.00054	-0.00052	-0.00070	-0.00036	-0.00051	-0.00061
5,544	-0.00073	-0.00055	-0.00070	-0.00067	-0.00060	-0.00048	-0.00073	-0.00040	-0.00049	-0.00069
6,048	-0.00056	-0.00049	-0.00055	-0.00060	-0.00043	-0.00044	-0.00066	-0.00024	-0.00042	-0.00053
6,552	-0.00067	-0.00049	-0.00058	-0.00053	-0.00048	-0.00038	-0.00072	-0.00031	-0.00040	-0.00062
7,056	-0.00066	-0.00062	-0.00052	-0.00068	-0.00047	-0.00043	-0.00073	-0.00033	-0.00046	-0.00067
7,560	-0.00062	-0.00055	-0.00059	-0.00064	-0.00051	-0.00051	-0.00078	-0.00031	-0.00046	-0.00076
8,064	-0.00052	-0.00051	-0.00049	-0.00060	-0.00043	-0.00042	-0.00060	-0.00026	-0.00040	-0.00066
8,568	-0.00043	-0.00052	-0.00038	-0.00057	-0.00036	-0.00038	-0.00065	-0.00019	-0.00041	-0.00056

Table D-4b. $\delta v'$ Relative Chromaticity Coordinates for the First Example LED.

$\delta v'$	Sample #									
Time, h	11	12	13	14	15	16	17	18	19	20
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
504	-0.00050	-0.00020	-0.00031	-0.00046	-0.00033	-0.00041	-0.00020	-0.00020	-0.00013	-0.00027
1,008	-0.00048	-0.00025	-0.00037	-0.00058	-0.00044	-0.00039	-0.00031	-0.00025	-0.00027	-0.00032
1,512	-0.00052	-0.00029	-0.00038	-0.00059	-0.00041	-0.00045	-0.00035	-0.00022	-0.00028	-0.00036
2,016	-0.00057	-0.00031	-0.00040	-0.00063	-0.00047	-0.00042	-0.00040	-0.00012	-0.00030	-0.00027
2,520	-0.00063	-0.00034	-0.00043	-0.00058	-0.00053	-0.00051	-0.00034	-0.00029	-0.00030	-0.00037
3,024	-0.00064	-0.00035	-0.00044	-0.00065	-0.00054	-0.00048	-0.00034	-0.00029	-0.00035	-0.00041
3,528	-0.00063	-0.00034	-0.00045	-0.00063	-0.00056	-0.00047	-0.00038	-0.00032	-0.00037	-0.00038
4,032	-0.00060	-0.00035	-0.00044	-0.00057	-0.00050	-0.00053	-0.00037	-0.00029	-0.00031	-0.00036
4,536	-0.00060	-0.00035	-0.00044	-0.00056	-0.00054	-0.00049	-0.00037	-0.00032	-0.00034	-0.00036
5,040	-0.00067	-0.00040	-0.00050	-0.00066	-0.00064	-0.00053	-0.00042	-0.00038	-0.00042	-0.00038
5,544	-0.00074	-0.00037	-0.00059	-0.00063	-0.00068	-0.00056	-0.00035	-0.00042	-0.00041	-0.00049
6,048	-0.00060	-0.00034	-0.00041	-0.00053	-0.00052	-0.00048	-0.00032	-0.00027	-0.00039	-0.00038
6,552	-0.00066	-0.00032	-0.00043	-0.00045	-0.00059	-0.00047	-0.00034	-0.00034	-0.00036	-0.00046
7,056	-0.00068	-0.00047	-0.00049	-0.00050	-0.00067	-0.00046	-0.00033	-0.00035	-0.00048	-0.00052
7,560	-0.00062	-0.00039	-0.00059	-0.00047	-0.00072	-0.00052	-0.00033	-0.00027	-0.00048	-0.00048
8,064	-0.00062	-0.00037	-0.00045	-0.00046	-0.00070	-0.00049	-0.00023	-0.00024	-0.00050	-0.00054
8,568	-0.00060	-0.00036	-0.00038	-0.00034	-0.00058	-0.00051	-0.00023	-0.00021	-0.00044	-0.00051

After calculating the relative chromaticity coordinates for each LED, the average relative chromaticity coordinates for each time period are calculated. This is shown in **Table D-5**.

Table D-5. Average Relative Chromaticity Coordinates for the First Example LED.

Time, h	Average	
	$\delta u'$	$\delta v'$
0	0.000000	0.000000
504	-0.000562	-0.000326
1,008	-0.000584	-0.000384
1,512	-0.000614	-0.000411
2,016	-0.000644	-0.000427
2,520	-0.000691	-0.000472
3,024	-0.000671	-0.000470
3,528	-0.000635	-0.000479
4,032	-0.000620	-0.000468
4,536	-0.000569	-0.000455
5,040	-0.000680	-0.000537
5,544	-0.000723	-0.000564
6,048	-0.000688	-0.000458
6,552	-0.000690	-0.000480
7,056	-0.000691	-0.000526
7,560	-0.000772	-0.000530
8,064	-0.000816	-0.000475
8,568	-0.000830	-0.000431

This is followed by calculating the differential chromaticities starting at approximately $t = 2,000$ hours:

$$\delta u'(t)^* = [u'(t) - u'(t - \Delta t)]/\Delta t = [\delta u'(t) - \delta u'(t - \Delta t)]/\Delta t$$

$$\delta v'(t)^* = [v'(t) - v'(t - \Delta t)]/\Delta t = [\delta v'(t) - \delta v'(t - \Delta t)]/\Delta t$$

$$\Delta t = t_k - t_{k-1}$$

It is important to note that the differential is calculated as a backwards difference. This uses the chromaticity data at 1,512 hours. The calculation method thus requires information from the time interval before the approximately 2,000-hour start time. In this example, although the relative chromaticities were calculated at 504 hours and 1,008 hours, these data points have no impact on the further calculations and may be omitted.

The results of this step are shown in **Table D-6**.

Table D-6. Differential Chromaticities for the First Example LED.

Time, h	$\delta u'^*$	$\delta v'^*$
0		
504		
1,008		
1,512		
2,016	-5.9524E-08	-3.1746E-08
2,520	-9.3254E-08	-8.9286E-08
3,024	3.9683E-08	3.9683E-09
3,528	7.1429E-08	-1.7857E-08
4,032	2.9762E-08	2.1825E-08
4,536	1.0119E-07	2.5794E-08
5,040	-2.2024E-07	-1.6270E-07
5,544	-8.5317E-08	-5.3571E-08
6,048	6.9444E-08	2.1032E-07
6,552	-3.9683E-09	-4.3651E-08
7,056	-1.9841E-09	-9.1270E-08
7,560	-1.6071E-07	-7.9365E-09
8,064	-8.7302E-08	1.0913E-07
8,568	-2.7778E-08	8.7302E-08

The differential chromaticities are then plotted against time, and a linear fit to the data is calculated. For a set of n experimental data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the least-squares straight line fit, where n is the total number of data points used for curve fitting according to **Section 5.2.5**, is:

$$y = mx + b ,$$

where m is the slope:

$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} ,$$

and b is the intercept:

$$b = \frac{\sum y - m\sum x}{n} ,$$

and where Σ stands for the sum of n terms:

$$\sum xy = x_1y_1 + x_2y_2 + \dots + x_ny_n$$

$$\sum x = x_1 + x_2 + \dots + x_n$$

$$\sum y = y_1 + y_2 + \dots + y_n$$

$$\sum x^2 = x_1^2 + x_2^2 + \dots + x_n^2$$

after substituting these terms:

$$x_k = t_k,$$

$$y_{ku'} = \delta u'^*_k,$$

$$y_{kv'} = \delta v'^*_k,$$

where $k = 1, 2, \dots, n$

These calculations are shown graphically in **Figures D-1** and **D-2**. Although the values in the preceding tables are using rounded values, the calculations should be carried out using as many significant figures as feasible.

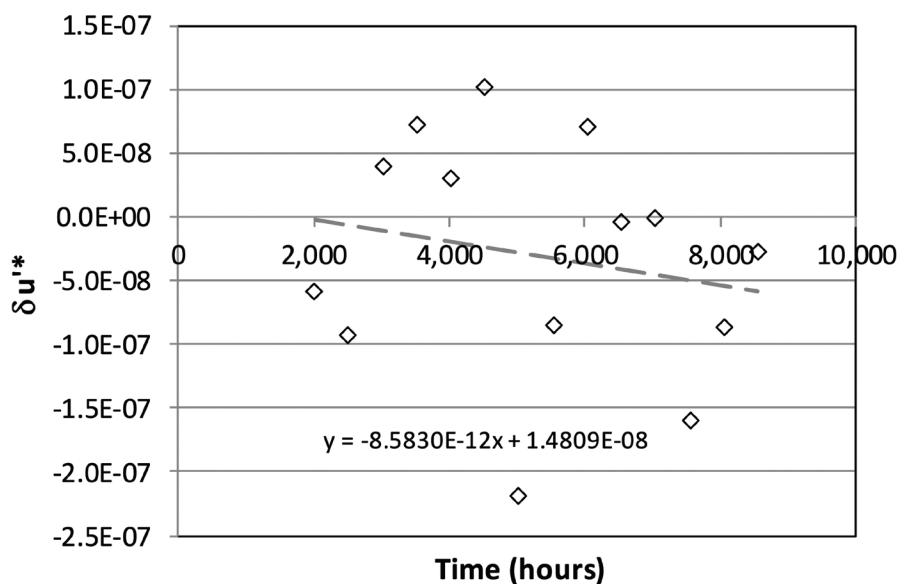


Figure D-1. Plot of $\delta u'^*$ data versus time and results of linear fit calculations for example LED.

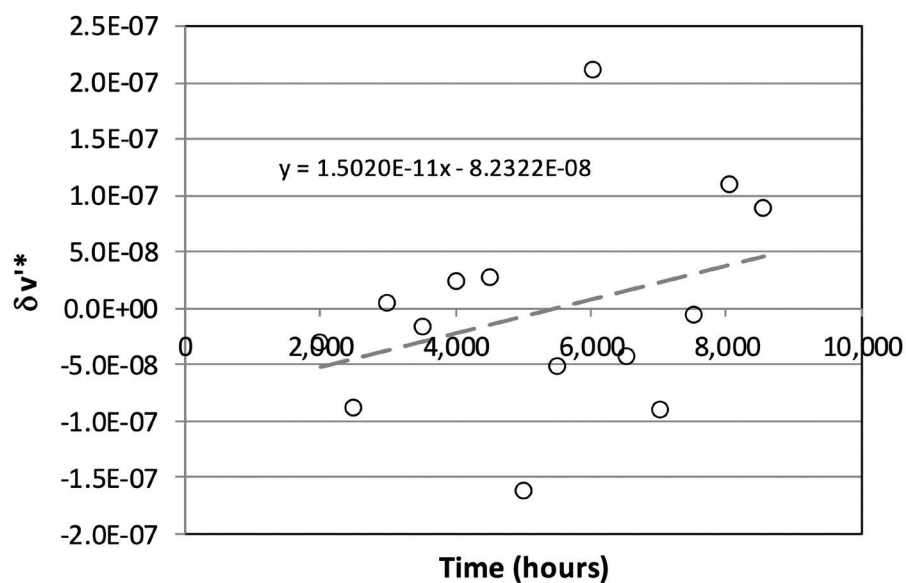


Figure D-2. Plot of $\delta v'^*$ data versus time and results of linear fit calculations for example LED.

The correlation coefficients for the data plotted in **Figure D-1** and **Figure D-2** are due to the extremely small changes in differential chromaticity. The differential chromaticities are used to calculate the behavior of the relative chromaticities which are in turn used to calculate $\Delta u'v'$.

When the differential chromaticity parameters are compared against the relative chromaticity values, it can be seen that they do a reasonable job of representing the actual data. If **Figure D-3**, the modeled data is based on the differential chromaticity parameters for $\delta u'^*$. While the differential chromaticity fit gave a correlation coefficient of $r^2 = 0.0381$, the modeled relative chromaticity versus the actual relative chromaticity ($\delta u'$) has a much higher correlation coefficient of $r^2 = 0.7922$.

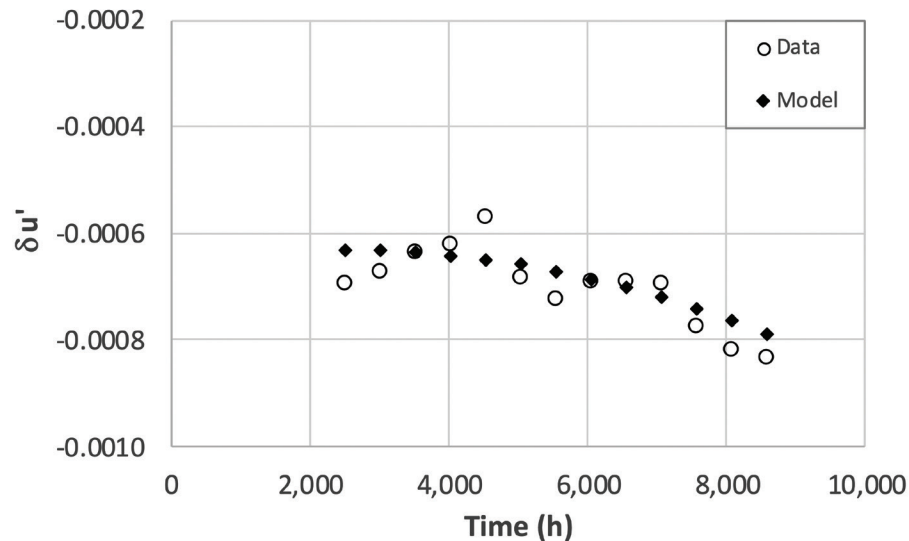


Figure D3. Comparison of the behavior of $\delta u'$ versus modeled behavior for first example LED.

A repeat the comparison for the behavior of $\delta v'$ against the modeled behavior from the parameters calculated for $\delta v'^*$ is shown in **Figure D-4**. It is noted that the parameters for $\delta v'^*$ had a correlation coefficient of $r^2 = 0.1146$. The modeled relative chromaticity versus the actual relative chromaticity ($\delta v'$) again has a higher correlation coefficient of $r^2 = 0.2350$.

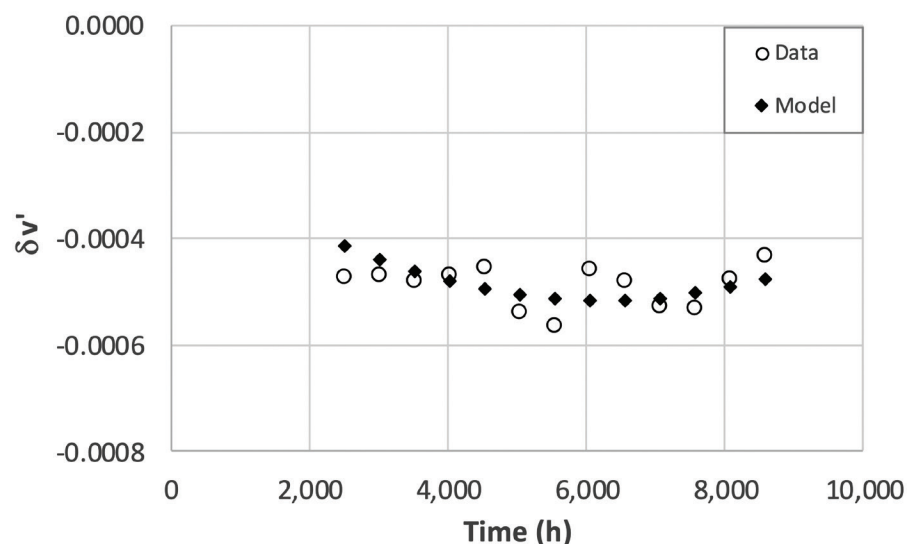


Figure D-4. Comparison of the behavior $\delta v'$ versus modeled behavior for first example LED.

Combining the modeled behavior for both relative chromaticity values allows a comparison against the chromaticity shift ($\Delta u'v'$). This is shown in **Figure D-5**. The correlation coefficient over the entire interval is $r^2 = 0.6357$.

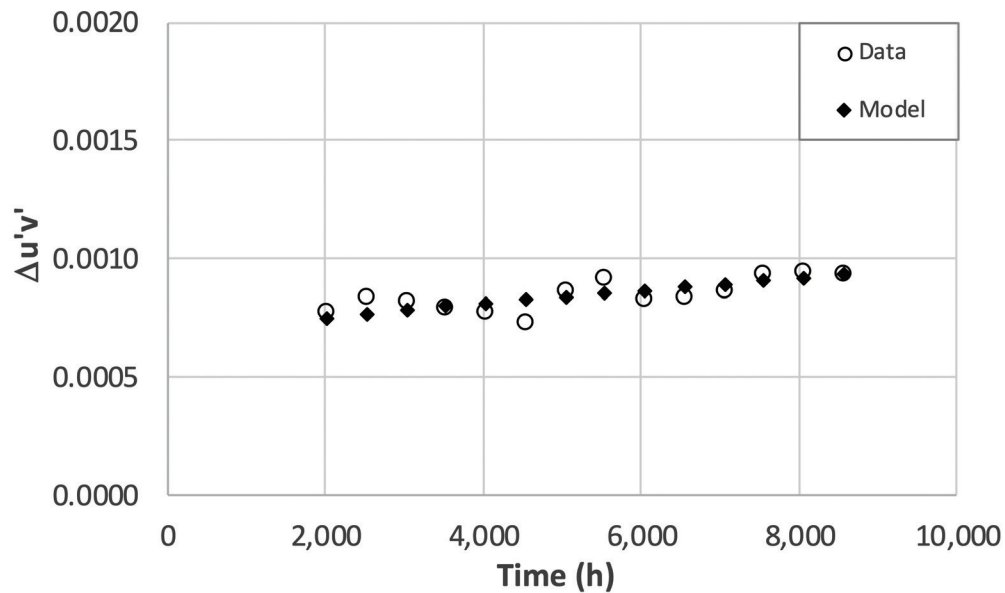


Figure D-5. Comparison of behavior of chromaticity shift ($\Delta u'v'$) versus modeled behavior for first example LED.

An initial report is shown in **Table D-7**. Chromaticity shift lifetimes for the first example LED are calculated in **Annex E**.

Table D-7. Initial TM-35 Report Data for First Example LED.

Sample LED	Value
Sample size	20
Number of failures	0
Test duration, hours	8,568
Measurement interval, hours	504
DUT drive current, mA	500
Tested case temperature, °C	105
Reference time, hours	8,568
Reference $\delta u'$	-0.000830
Reference $\delta v'$	-0.000431
$\delta u'^*$ slope, hr^{-2}	-8.6325E-12
$\delta u'^*$ intercept, hr^{-1}	1.5084E-08
$\delta v'^*$ slope, hr^{-2}	1.5020E-11
$\delta v'^*$ intercept, hr^{-1}	-8.2322E-08

A second example is provided to further assist users to make a comparison with their own calculations. The operating conditions for the second example LED were a case temperature of 85 °C and a drive current of 1,000 mA. **Tables D-8a,b** and **D-9a,b** provide example u' and v' data, respectively, for the second example LED.

Table D-8a. u' Data for Second Example LED.

CIE u'	Sample #									
Time, h	1	2	3	4	5	6	7	8	9	10
0	0.24680	0.24604	0.24549	0.24675	0.24620	0.24610	0.24610	0.24607	0.24778	0.24651
1,000	0.24628	0.24555	0.24489	0.24622	0.24566	0.24562	0.24566	0.24555	0.24723	0.24590
2,000	0.24620	0.24546	0.24485	0.24614	0.24561	0.24554	0.24562	0.24551	0.24717	0.24578
3,000	0.24617	0.24538	0.24480	0.24607	0.24557	0.24551	0.24556	0.24544	0.24709	0.24575
4,000	0.24616	0.24548	0.24485	0.24609	0.24555	0.24552	0.24559	0.24551	0.24712	0.24581
5,000	0.24612	0.24546	0.24480	0.24604	0.24553	0.24556	0.24557	0.24550	0.24710	0.24575
6,000	0.24623	0.24555	0.24485	0.24617	0.24565	0.24561	0.24567	0.24563	0.24722	0.24586
7,000	0.24618	0.24540	0.24487	0.24608	0.24571	0.24543	0.24547	0.24560	0.24715	0.24583

Table D-8b. u' Data for Second Example LED.

CIE u'	Sample #									
Time, h	11	12	13	14	15	16	17	18	19	20
0	0.24556	0.24480	0.24645	0.24773	0.24708	0.24647	0.24523	0.24640	0.24775	0.24410
1,000	0.24509	0.24436	0.24593	0.24720	0.24664	0.24605	0.24478	0.24590	0.24715	0.24352
2,000	0.24500	0.24425	0.24584	0.24708	0.24655	0.24596	0.24471	0.24583	0.24705	0.24344
3,000	0.24496	0.24418	0.24582	0.24706	0.24657	0.24592	0.24468	0.24580	0.24702	0.24339
4,000	0.24496	0.24425	0.24583	0.24710	0.24659	0.24596	0.24467	0.24583	0.24706	0.24339
5,000	0.24495	0.24427	0.24576	0.24708	0.24658	0.24597	0.24469	0.24577	0.24708	0.24338
6,000	0.24502	0.24426	0.24583	0.24711	0.24666	0.24586	0.24450	0.24590	0.24701	0.24340
7,000	0.24494	0.24421	0.24579	0.24708	0.24661	0.24577	0.24462	0.24587	0.24701	0.24336

Table D-9a. v' Data for Second Example LED.

CIE v'	Sample #									
Time, h	1	2	3	4	5	6	7	8	9	10
0	0.52085	0.52155	0.51707	0.51780	0.52177	0.51793	0.51981	0.52282	0.52260	0.52224
1,000	0.52026	0.52126	0.51655	0.51716	0.52139	0.51748	0.51946	0.52257	0.52224	0.52173
2,000	0.52024	0.52120	0.51649	0.51707	0.52128	0.51742	0.51935	0.52247	0.52218	0.52166
3,000	0.52016	0.52113	0.51638	0.51697	0.52116	0.51731	0.51934	0.52243	0.52207	0.52161
4,000	0.52011	0.52106	0.51625	0.51693	0.52110	0.51722	0.51928	0.52227	0.52201	0.52149
5,000	0.52002	0.52106	0.51620	0.51698	0.52097	0.51715	0.51925	0.52218	0.52194	0.52138
6,000	0.51995	0.52095	0.51623	0.51687	0.52094	0.51700	0.51925	0.52216	0.52187	0.52138
7,000	0.52006	0.52086	0.51640	0.51690	0.52103	0.51693	0.51919	0.52222	0.52187	0.52147

Table D-9b. v' Data for Second Example LED.

CIE v'	Sample #									
Time, h	11	12	13	14	15	16	17	18	19	20
0	0.52171	0.52095	0.51752	0.51846	0.52145	0.51646	0.51910	0.52389	0.52429	0.52085
1,000	0.52142	0.52052	0.51703	0.51795	0.52113	0.51596	0.51867	0.52362	0.52395	0.52038
2,000	0.52141	0.52043	0.51695	0.51783	0.52099	0.51585	0.51863	0.52350	0.52389	0.52039
3,000	0.52125	0.52035	0.51684	0.51774	0.52091	0.51573	0.51851	0.52346	0.52382	0.52031
4,000	0.52124	0.52027	0.51673	0.51770	0.52080	0.51566	0.51847	0.52333	0.52377	0.52016
5,000	0.52115	0.52020	0.51658	0.51767	0.52075	0.51562	0.51844	0.52322	0.52365	0.51999
6,000	0.52114	0.52013	0.51661	0.51768	0.52078	0.51542	0.51818	0.52314	0.52356	0.51992
7,000	0.52102	0.52007	0.51661	0.51773	0.52073	0.51537	0.51832	0.52309	0.52354	0.51991

The first step in the calculations is calculating the relative chromaticity coordinates for the each LED at each time interval. The relative chromaticity coordinates are calculated using:

$$\delta u'(t)_i = u'(t)_i - u'(0)_i$$

$$\delta v'(t)_i = v'(t)_i - v'(0)_i$$

This is shown in **Tables D-10a,b** and **D-11a,b**.

Table D-10a. $\delta u'$ Relative Chromaticity Coordinates for the Second Example LED.

$\delta u'$	Sample #									
Time, h	1	2	3	4	5	6	7	8	9	10
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
1,000	-0.00052	-0.00049	-0.00060	-0.00053	-0.00054	-0.00048	-0.00044	-0.00052	-0.00055	-0.00061
2,000	-0.00060	-0.00058	-0.00064	-0.00061	-0.00059	-0.00056	-0.00048	-0.00056	-0.00061	-0.00073
3,000	-0.00063	-0.00066	-0.00069	-0.00068	-0.00063	-0.00059	-0.00054	-0.00063	-0.00069	-0.00076
4,000	-0.00064	-0.00056	-0.00064	-0.00066	-0.00065	-0.00058	-0.00051	-0.00056	-0.00066	-0.00070
5,000	-0.00068	-0.00058	-0.00069	-0.00071	-0.00067	-0.00054	-0.00053	-0.00057	-0.00068	-0.00076
6,000	-0.00057	-0.00049	-0.00064	-0.00058	-0.00055	-0.00049	-0.00043	-0.00044	-0.00056	-0.00065
7,000	-0.00062	-0.00064	-0.00062	-0.00067	-0.00049	-0.00067	-0.00063	-0.00047	-0.00063	-0.00068

Table D-10b. $\delta u'$ Relative Chromaticity Coordinates for the Second Sample LED.

$\delta u'$	Sample #									
Time, h	11	12	13	14	15	16	17	18	19	20
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
1,000	-0.00047	-0.00044	-0.00052	-0.00053	-0.00044	-0.00042	-0.00045	-0.00050	-0.00060	-0.00058
2,000	-0.00056	-0.00055	-0.00061	-0.00065	-0.00053	-0.00051	-0.00052	-0.00057	-0.00070	-0.00066
3,000	-0.00060	-0.00062	-0.00063	-0.00067	-0.00051	-0.00055	-0.00055	-0.00060	-0.00073	-0.00071
4,000	-0.00060	-0.00055	-0.00062	-0.00063	-0.00049	-0.00051	-0.00056	-0.00057	-0.00069	-0.00071
5,000	-0.00061	-0.00053	-0.00069	-0.00065	-0.00050	-0.00050	-0.00054	-0.00063	-0.00067	-0.00072
6,000	-0.00054	-0.00054	-0.00062	-0.00062	-0.00042	-0.00061	-0.00073	-0.00050	-0.00074	-0.00070
7,000	-0.00062	-0.00059	-0.00066	-0.00065	-0.00047	-0.00070	-0.00061	-0.00053	-0.00074	-0.00074

Table D-11a. $\delta v'$ Relative Chromaticity Coordinates for the Second Sample LED.

$\delta v'$	Sample #									
Time, h	1	2	3	4	5	6	7	8	9	10
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
1,000	-0.00059	-0.00029	-0.00052	-0.00064	-0.00038	-0.00045	-0.00035	-0.00025	-0.00036	-0.00051
2,000	-0.00061	-0.00035	-0.00058	-0.00073	-0.00049	-0.00051	-0.00046	-0.00035	-0.00042	-0.00058
3,000	-0.00069	-0.00042	-0.00069	-0.00083	-0.00061	-0.00062	-0.00047	-0.00039	-0.00053	-0.00063
4,000	-0.00074	-0.00049	-0.00082	-0.00087	-0.00067	-0.00071	-0.00053	-0.00055	-0.00059	-0.00075
5,000	-0.00083	-0.00049	-0.00087	-0.00082	-0.00080	-0.00078	-0.00056	-0.00064	-0.00066	-0.00086
6,000	-0.00090	-0.00060	-0.00084	-0.00093	-0.00083	-0.00093	-0.00056	-0.00066	-0.00073	-0.00086
7,000	-0.00079	-0.00069	-0.00067	-0.00090	-0.00074	-0.00100	-0.00062	-0.00060	-0.00073	-0.00077

Table D-11b. $\delta v'$ Relative Chromaticity Coordinates for Second Example LED.

$\delta v'$	Sample #									
Time, h	11	12	13	14	15	16	17	18	19	20
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
1,000	-0.00029	-0.00043	-0.00049	-0.00051	-0.00032	-0.00050	-0.00043	-0.00027	-0.00034	-0.00047
2,000	-0.00030	-0.00052	-0.00057	-0.00063	-0.00046	-0.00061	-0.00047	-0.00039	-0.00040	-0.00046
3,000	-0.00046	-0.00060	-0.00068	-0.00072	-0.00054	-0.00073	-0.00059	-0.00043	-0.00047	-0.00054
4,000	-0.00047	-0.00068	-0.00079	-0.00076	-0.00065	-0.00080	-0.00063	-0.00056	-0.00052	-0.00069
5,000	-0.00056	-0.00075	-0.00094	-0.00079	-0.00070	-0.00084	-0.00066	-0.00067	-0.00064	-0.00086
6,000	-0.00057	-0.00082	-0.00091	-0.00078	-0.00067	-0.00104	-0.00092	-0.00075	-0.00073	-0.00093
7,000	-0.00069	-0.00088	-0.00091	-0.00073	-0.00072	-0.00109	-0.00078	-0.00080	-0.00075	-0.00094

After calculating the relative chromaticity coordinates for each LED, the averaged relative chromaticity coordinates for each time period are calculated. This is shown in **Table D-12**.

Table D-12. Average Relative Chromaticity Data for Second Example LED.

Time, h	Average	
	$\delta u'$	$\delta v'$
0	0.000000	0.000000
1,000	-0.000512	-0.000420
2,000	-0.000591	-0.000495
3,000	-0.000634	-0.000582
4,000	-0.000605	-0.000664
5,000	-0.000623	-0.000736
6,000	-0.000571	-0.000798
7,000	-0.000622	-0.000790

This is followed by calculating the differential chromaticities starting at approximately $t = 2,000$ hours:

$$\delta u'(t)^* = [u'(t) - u'(t - \Delta t)]/\Delta t = [\delta u'(t) - \delta u'(t - \Delta t)]/\Delta t$$

$$\delta v'(t)^* = [v'(t) - v'(t - \Delta t)]/\Delta t = [\delta v'(t) - \delta v'(t - \Delta t)]/\Delta t$$

$$\Delta t = t_k - t_{k-1}$$

The results of this step are shown in **Table D-13**.

Table D-13. Differential Chromaticities for Second Example LED.

Time, h	Average	
	$\delta u'$	$\delta v'$
0		
1,000		
2,000	-7.9500E-08	-7.5000E-08
3,000	-4.2500E-08	-8.7500E-08
4,000	2.9000E-08	-8.1500E-08
5,000	-1.8000E-08	-7.2500E-08
6,000	5.1500E-08	-6.2000E-08
7,000	-5.0500E-08	8.0000E-09

The differential chromaticities are then plotted against time, and a linear fit to the data is calculated. For a set of n experimental data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the least-squares straight line fit, where n is the total number of data points used for curve fitting according to **Section 5.2.5**, is:

$$y = mx + b,$$

where m is the slope:

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2},$$

and b is the intercept:

$$b = \frac{\sum y - m \sum x}{n},$$

and where Σ stands for the sum of n terms:

$$\sum xy = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

$$\sum x = x_1 + x_2 + \dots + x_n$$

$$\sum y = y_1 + y_2 + \dots + y_n$$

$$\sum x^2 = x_1^2 + x_2^2 + \dots + x_n^2$$

after substituting these terms:

$$x_k = t_k,$$

$$y_{ku'} = \delta u'^*_{k'},$$

$$y_{kv'} = \delta v'^*_{k'},$$

where $k = 1, 2, \dots, n$

The results of these calculations are shown graphically in **Figures D-6** and **D-7**.

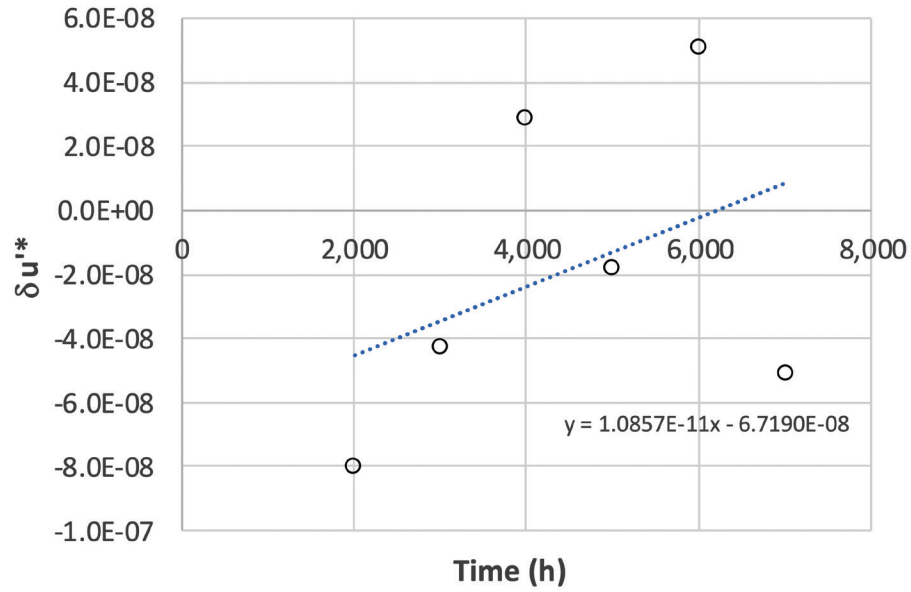


Figure D-6. Plot of $\delta u'^*$ data versus time and results of linear fit calculations for example LED.

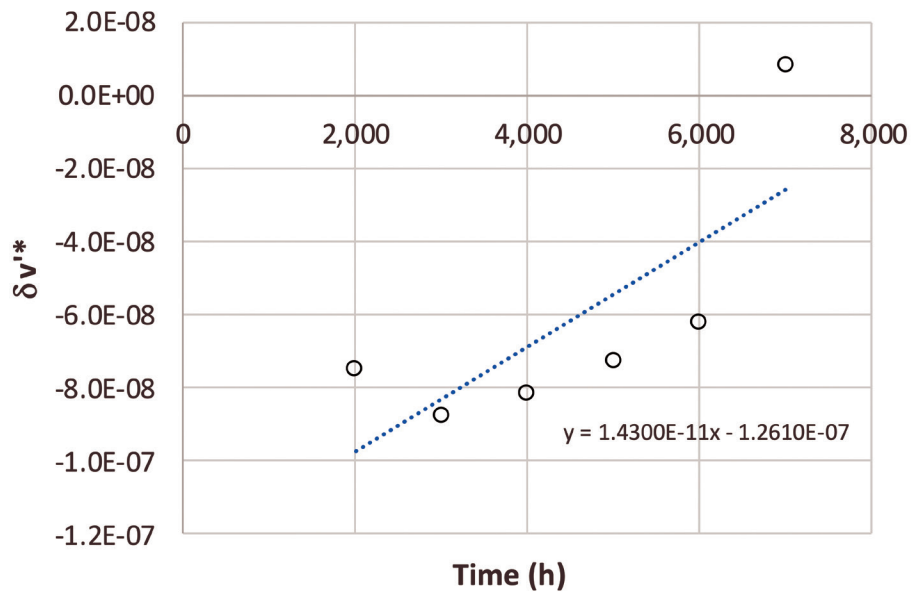


Figure D-7. Plot of $\delta v'^*$ data versus time and results of linear fit calculations for example LED.

An initial report is shown in **Table D-14**. Chromaticity shift lifetimes for the second example LED are calculated in **Annex E**.

Table D-14. Initial TM-35 Report Data for Second Example LED

Sample LED	Value
Sample size	20
Number of failures	0
Test duration, hours	1,000
Measurement interval, hours	85
DUT drive current, mA	7,000
Tested case temperature, °C	1,000
Reference time, hours	7,000
Reference $\delta u'$	-0.00062
Reference $\delta v'$	-0.00079
$\delta u'^*$ slope, hr ⁻²	1.0857E-11
$\delta u'^*$ intercept, hr ⁻¹	-6.7191E-8
$\delta v'^*$ slope, hr ⁻²	1.4300E-11
$\delta v'^*$ intercept, hr ⁻¹	-1.2610E-07

Annex E – Extrapolation Methods

Extrapolating chromaticity shift behavior forward in time is a matter of numerical integration. For this procedure, the relative chromaticity coordinates for the final time interval used in the linear-fit calculations are used as a reference point.

The values for the differential chromaticities are used along with the average time interval of the dataset. The first step is to enter the time and relative chromaticities of the final data point in the test set into a table, as shown in **Table E-1**.

Table E-1. Initial Values in the Chromaticity Shift Extrapolation Table for First Example LED

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
8,568	-0.000830	-0.000431	0.00093523		

Differential chromaticity analysis (DCA; see **Annex C**) assumes that differential chromaticities are a function of time. Therefore, using the slopes and intercepts for each differential chromaticity, it is then possible to calculate the differential chromaticity for any arbitrary time in the future. Thus,

$$\delta u'^*(t)\Delta t = (au't + bu')\Delta t$$

becomes:

$$\delta u'^*(8,568) = [(-8.5830 \times 10^{-12} \text{ h}^{-2})(8568 \text{ h}) + 1.4809 \times 10^{-8} \text{ h}^{-1}]\Delta t$$

For this dataset, Δt is constant at 504 hours:

$$\delta u'^*(8,568) = [(-8.5830 \times 10^{-12} \text{ h}^{-2} \times 8568 \text{ h}) + 1.4809 \times 10^{-8} \text{ h}^{-1}] \times 504 \text{ h}$$

$$\delta u'^*(8,568) = (-7.3539 \times 10^{-8} \text{ h}^{-1} + 1.4809 \times 10^{-8} \text{ h}^{-1}) \times 504 \text{ h}$$

$$\delta u'^*(8,568) = -2.9600 \times 10^{-5}$$

Similarly,

$$\delta v'^*(t) = (av't + bv')\Delta t$$

becomes:

$$\delta v'^*(8,568) = [(1.5020 \times 10^{-11} \text{ h}^{-2} \times 8,568 \text{ h}) - 8.2322 \times 10^{-8} \text{ h}^{-1}] \times 504 \text{ h}$$

$$\delta v'^*(8,568) = (1.2860 \times 10^{-7} \text{ h}^{-1} - 8.2322 \times 10^{-8} \text{ h}^{-1}) \times 504 \text{ h}$$

$$\delta v'^*(8,568) = 2.3370 \times 10^{-5}$$

This allows the first row in the chromaticity shift extrapolation table to be completed (see **Table E-2**).

The first three columns of the second row of the chromaticity shift extrapolation table may now be populated. The time is incremented by the average Δt for the test dataset, and new values of $\delta u'$ and $\delta v'$ are calculated:

Table E-2. Completed First Row for the Chromaticity Shift Extrapolation Table for First Example LED

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
8,568	-0.000830	-0.000431	0.00093523	-2.9600E-05	2.3370E-05

$$\delta u'(t + \Delta t) = \delta u'(t) + \delta u'^*(t)$$

$$\delta u'(8,568+504) = \delta u'(8,568) + \delta u'^*(8,568)$$

$$\delta u'(9,072) = -0.000830 - 2.9600 \times 10^{-5} = -0.000860$$

Similarly:

$$\delta v'(t + \Delta t) = \delta v'(t) + \delta v'^*(t)$$

$$\delta v'(8,568+504) = \delta v'(8,568) + \delta v'^*(8,568)$$

$$\delta v'(9,072) = -0.000431 + 2.3370 \times 10^{-5} = -0.000431$$

This also allows calculation of the predicted value of $\Delta u'v'$:

$$\Delta u'v'_t = \sqrt{[\delta u'(t)]^2 + [\delta v'(t)]^2}$$

$$\Delta u'v'_{9,072} = \sqrt{[-0.000860]^2 + [-0.000431]^2} = 0.00095135$$

Table E-3 shows the new chromaticity shift extrapolation data.

Table E-3. Chromaticity Shift Extrapolation Table after Calculations for the First Time Step for First Example LED

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
8,568	-0.000830	-0.000431	0.00093523	-2.9600E-05	2.3370E-05
9,072	-0.000860	-0.000408	0.00095135		

Repeating this process for the second time step gives the results provided in **Table E-4**.

Table E-4. Chromaticity Shift Extrapolation Table after Calculations for the Second Time Step for First Example LED

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
8,568	-0.000830	-0.000431	0.00093523	-2.9600E-05	2.3370E-05
9,072	-0.000860	-0.000408	0.00095135	-3.1780E-05	2.7185E-05
9,576	-0.000891	-0.000380	0.00096917		

This dataset contained 20 DUTs, extended to 8,568 hours, and had an average interval between measurements of 504 hours. Per **Section 5.2.7**, this allows an extrapolation limit of 6x, or 51,408 hours. Thus, the process may be continued until the time reaches the chromaticity shift life or this time.

The projected behavior for the first example LED is shown graphically in **Figure E-1**. The time scale was truncated at 35,000 hours, since the CS7 is projected to happen at an earlier time.

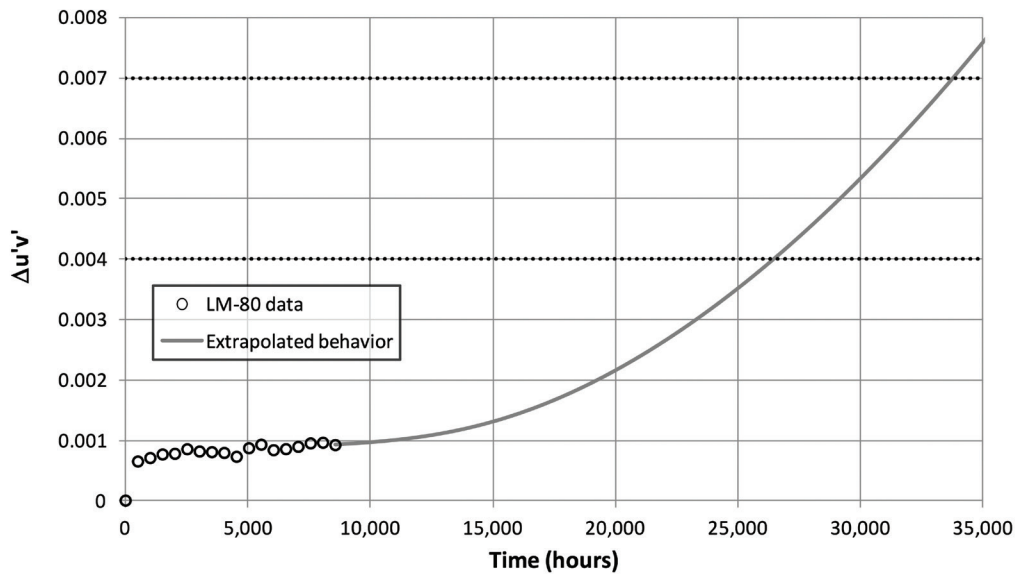


Figure E-1. Projected chromaticity shift behavior for the first example LED. LM-80 data points are included for reference. Horizontal dotted lines indicate $\Delta u'v' = 0.004$ and $\Delta u'v' = 0.007$.

As seen in **Table E-5**, CS4 is bracketed by the entries for 26,208 hours and 26,712 hours. Linear behavior is assumed between these time data points. The slope and intercept are calculated by:

$$y = mx + b$$

$$m = \frac{\Delta u'v'_2 - \Delta u'v'_1}{t_2 - t_1} = \frac{0.00413111 - 0.00395354}{26,208 - 25,704} = \frac{0.00017757}{504} = 3.5253 \times 10^{-7}$$

$$b = \Delta u'v'_1 - mt_1 = 0.00395354 - 3.5253 \times 10^{-7} \times 25,704 = -0.0051025$$

$$t_{CS4} = \frac{0.004 - b}{m} = \frac{0.004 + 0.0051025}{3.5253 \times 10^{-7}} = 25,836 \text{ hours}$$

Rounded to the nearest 100 hours, CS4 = 25,800 hours.

Table E-5. Chromaticity Shift Extrapolation Table for First Example LED Showing Entries in the Vicinity of CS4 and CS7

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
25,200	-0.002958	0.002355	0.00378076	-1.0155E-04	1.4928E-04
25,704	-0.003060	0.002504	0.00395354	-1.0373E-04	1.5309E-04
26,208	-0.003163	0.002657	0.00413111	-1.0591E-04	1.5691E-04
26,712	-0.003269	0.002814	0.00431344	-1.0809E-04	1.6072E-04
...	...	...	...	...	...
32,256	-0.004578	0.004792	0.00662716	-1.3207E-04	2.0269E-04
32,760	-0.004710	0.004994	0.00686509	-1.3425E-04	2.0651E-04
33,264	-0.004844	0.005201	0.00710757	-1.3643E-04	2.1032E-04
33,768	-0.004981	0.005411	0.00735459	-1.3861E-04	2.1414E-04

Following the same procedure, the time to reach CS7 is calculated as 33,040 hours, which is rounded to 33,000 hours. With these calculations, the table portion of the TM-35 Report can now be completed (see **Table E-6**).

Table E-6. Full TM-35 Report for First Example LED

Sample LED	Value
Sample size	20
Number of failures	0
Test duration, hours	8,568
Measurement interval, hours	504
DUT drive current, mA	500
Tested case temperature, °C	105
Reference time, hours	8,568
Reference $\delta u'$	-0.000830
Reference $\delta v'$	-0.000431
$\delta u'^*$ slope, hr ⁻²	-8.5830E-12
$\delta u'^*$ intercept, hr ⁻¹	1.4809E-08
$\delta v'^*$ slope, hr ⁻²	1.5020E-11
$\delta v'^*$ intercept, hr ⁻¹	-8.2322E-08
Calculated CS4, hours	25,836
Reported CS4(8.5k), hours	25,800
Calculated CS7, hours	33,040
Calculated CS7(8.5k), hours	33,000

The values for the $\delta u'$ and $\delta v'$ can be calculated using the numerical extrapolation and interpolation procedure above to calculate the values needed for the vector graphic (see **Table E-7** and **Figure E-2**).

Table E-7. Numerical Extrapolation Values of Differential Chromaticities at CS4 and CS7 for First Example LED

	$\delta u'$	$\delta v'$
CS4	-0.003085	0.002541
CS7	-0.004785	0.005109

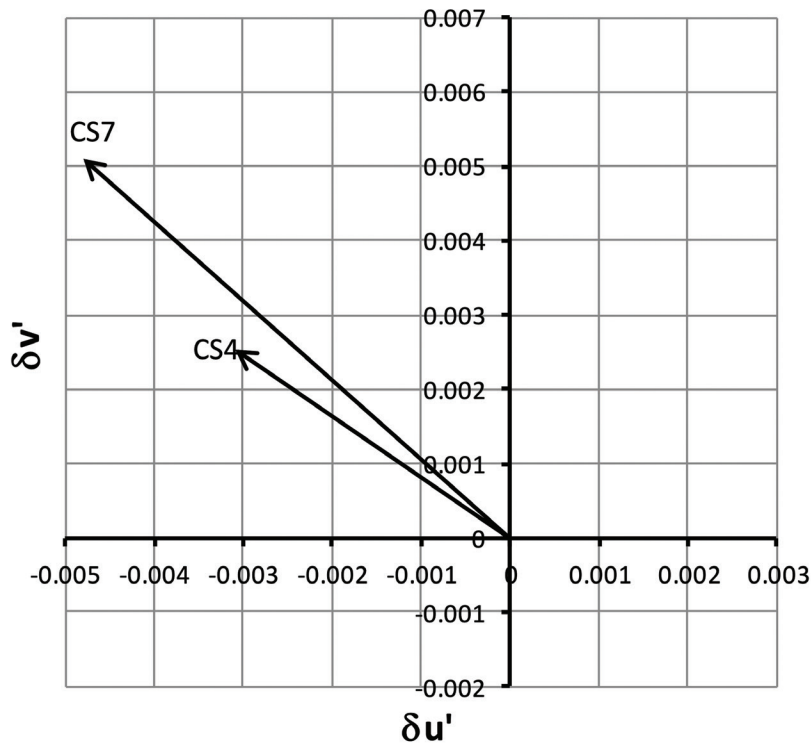


Figure E-2. Vector graphic for first example LED.

Repeating the process again for the second example LED, a table is created using the data for the final time interval. This data includes the elapsed time, the average relative chromaticity values ($\delta u'$, $\delta v'$) at the final time interval and the chromaticity shift ($\Delta u'v'$) is shown in **Table E-8**.

Table E-8. Initial Values in the Chromaticity Shift Extrapolation Table for the Second Example LED

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
7,000	-0.000622	-0.000790	0.0010052		

As stated previously, differential chromaticity analysis (DCA; see **Annex C**) assumes that differential chromaticities are a function of time. This allows the differential chromaticity values to be calculated for any arbitrary time using the slope and intercept for each differential chromaticity. Thus,

$$\delta u'^*(t)\Delta t = (a_u t + b_u)\Delta t$$

becomes:

$$\delta u'^*(7,000) = [(1.0857 \times 10^{-11} \text{ h}^{-2})(7000 \text{ h}) - 6.7191 \times 10^{-8} \text{ h}^{-1}]\Delta t$$

For this dataset, Δt is constant at 1,000 hours:

$$\begin{aligned}\delta u'^*(7,000) &= [(1.0857 \times 10^{-11} \text{ h}^{-2} \times 7000 \text{ h}) - 6.7191 \times 10^{-8} \text{ h}^{-1}] \times 1,000 \text{ h} \\ \delta u'^*(7,000) &= (7.600 \times 10^{-8} \text{ h}^{-1} - 6.7191 \times 10^{-8} \text{ h}^{-1}) \times 1,000 \text{ h} \\ \delta u'^*(7,000) &= 8.8080 \times 10^{-9}\end{aligned}$$

Similarly,

$$\delta v'^*(t) = (a_v t + b_v) \Delta t$$

becomes:

$$\begin{aligned}\delta v'^*(7,000) &= [(1.4300 \times 10^{-11} \text{ h}^{-2} \times 7000 \text{ h}) - 1.2610 \times 10^{-7} \text{ h}^{-1}] \times 1,000 \text{ h} \\ \delta v'^*(7,000) &= (1.0010 \times 10^{-7} \text{ h}^{-1} - 1.2610 \times 10^{-7} \text{ h}^{-1}) \times 1,000 \text{ h} \\ \delta v'^*(7,000) &= -2.6000 \times 10^{-5}\end{aligned}$$

This allows the first row in the chromaticity shift extrapolation table to be completed (see **Table E-9**).

Table E-9. Completed First Row for the Chromaticity Shift Extrapolation Table for First Example LED

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
7,000	-0.000622	-0.000790	0.0010052	8.8090E-06	-2.6000E-05

The first three columns of the second row of the chromaticity shift extrapolation table may now be populated. The time is incremented by the average Δt for the test dataset, and new values of $\delta u'$ and $\delta v'$ are calculated:

$$\begin{aligned}\delta u'(t + \Delta t) &= \delta u'(t) + \delta u'^*(t) \\ \delta u'(7,000 + 1,000) &= \delta u'(7,000) + \delta u'^*(7,000) \\ \delta u'(8,000) &= -0.000622 + 8.8090 \times 10^{-6} = -0.000613\end{aligned}$$

Similarly:

$$\begin{aligned}\delta v'(t + \Delta t) &= \delta v'(t) + \delta v'^*(t) \\ \delta v'(7,000 + 1,000) &= \delta v'(7,000) + \delta v'^*(7,000) \\ \delta v'(8,000) &= -0.000790 - 2.6000 \times 10^{-5} = -0.000819\end{aligned}$$

This also allows calculation of the predicted value of $\Delta u'v'$:

$$\begin{aligned}\Delta u'v'_t &= \sqrt{[\delta u'(t)]^2 + [\delta v'(t)]^2} \\ \Delta u'v'_{8,000} &= \sqrt{[-0.000613]^2 + [-0.000819]^2} = 0.0010204\end{aligned}$$

Table E-10 shows the new chromaticity shift extrapolation data.

Table E-10. Chromaticity Shift Extrapolation Table after Calculations for the First Time Step for the Second Example LED

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
7,000	-0.000622	-0.000790	0.0010052	8.8090E-06	-2.6000E-05
8,000	-0.000613	-0.000816	0.0010204		

Repeating this process for the second time step gives the results provided in **Table E-11**.

Table E-11. Chromaticity Shift Extrapolation Table after Calculations for the Second Time Step for the Second Example LED

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
7,000	-0.000622	-0.000790	0.0010052	8.8090E-06	-2.6000E-05
8,000	-0.000613	-0.000816	0.0010204	1.9666E-05	-1.1700E-05
9,000	-0.000593	-0.000828	0.0010182		

This dataset contained 20 DTUs, extended to 7,000 hours, and had an average interval between measurements of 1,000 hours. Per **Section 5.2.7**, this allows an extrapolation limit of 5.0x or 35,000 hours. Thus, the process may be continued until the time reaches the chromaticity shift life or this time.

The projected behavior for the second example LED is shown graphically in **Figure E-3**. The time scale was truncated at 35,000 hours, since this is the extrapolation limit for this dataset.

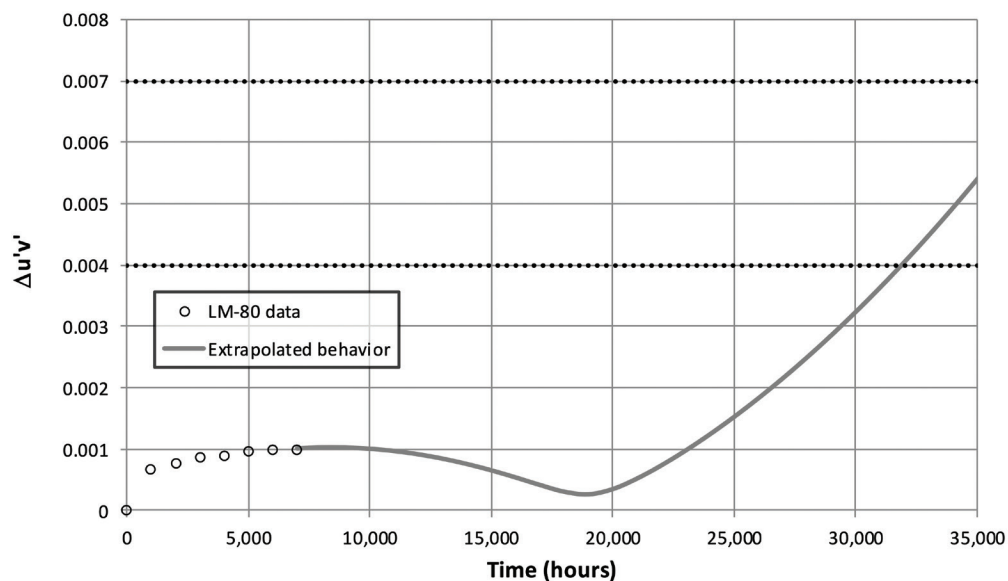


Figure E-3. Projected chromaticity shift behavior for the second example LED. LM-80 data points are included for reference. Horizontal dotted lines indicate $\Delta u'v' = 0.004$ and 0.007.

As seen in **Table E-12**, CS4 is bracketed by the entries for 31,000 hours and 32,000 hours. Linear behavior is assumed between these time data points. The slope and intercept are calculated by:

$$y = mx + b$$

$$m = \frac{\Delta u'v'_2 - \Delta u'v'_1}{t_2 - t_1} = \frac{0.00403463 - 0.00362005}{32,000 - 31,000} = \frac{0.00041458}{1,000} = 4.1458 \times 10^{-7}$$

$$b = \Delta u'v'_1 - mt_1 = 0.00362005 - 4.1458 \times 10^{-7} \times 31,000 = -0.0092318$$

$$t_{CS4} = \frac{0.004 - b}{m} = \frac{0.004 + 0.0092318}{4.1458 \times 10^{-7}} = 31,916 \text{ hours}$$

Rounded to the nearest 100 hours, CS4 = 31,900 hours.

Table E-12. Chromaticity Shift Extrapolation Table for the Second Example LED showing Entries in the Vicinity of CS4 and CS7

Time, h	$\delta u'$	$\delta v'$	$\Delta u'v'$	$\delta u'^*$	$\delta v'^*$
30,000	0.002328	0.002230	0.00322362	2.5852E-04	3.0290E-04
31,000	0.002586	0.002533	0.00362005	2.6938E-04	3.1720E-04
32,000	0.002856	0.002850	0.00403463	2.8023E-04	3.3150E-04
33,000	0.003136	0.003182	0.00446730	2.9109E-04	3.4580E-04
...	...	...	...	...	...
37,000	0.004366	0.004651	0.00637850	3.3452E-04	4.0300E-04
38,000	0.004700	0.005054	0.00690135	3.4538E-04	4.1730E-04
39,000	0.005045	0.005471	0.00744220	3.5623E-04	4.3160E-04
40,000	0.005402	0.005902	0.00800104	3.6709E-04	4.4590E-04

Following the same procedure, the time to reach CS7 is calculated as 38,182 hours, which exceeds the allowable extrapolation limit for this dataset. CS7 shall therefore be reported as CS7 > 35,000 hours.

With these calculations, the table portion of the TM-35 Report can now be completed (see **Table E-13**).

Table E-13. Full TM-35 Report for the Second Example LED

Sample LED	Value
Sample size	20
Number of failures	0
DUT drive current, mA	1,000
Tested case temperature, °C	85
Test duration, hours	8,000
Measurement interval, hours	1,000
Reference time, hours	8,000
Reference $\delta u'$	-0.000622
Reference $\delta v'$	-0.000790
$\delta u'^*$ slope, hr ⁻²	1.0857E-11
$\delta u'^*$ intercept, hr ⁻¹	-6.7191E-08
$\delta v'^*$ slope, hr ⁻²	1.4300E-11
$\delta v'^*$ intercept, hr ⁻¹	-1.2610E-07
Calculated CS4, hours	31,916
Reported CS4(7k), hours	32,000
Calculated CS7, hours	38,182
Reported CS7(7k), hours	>35,000

The values for $\delta u'$ and $\delta v'$ can be calculated using the numerical extrapolation and interpolation procedure above to calculate the values needed for the vector graphic (see **Table E-14** and **Figure E-4**). Since CS7 exceeds the allowable extrapolation limits for this dataset, no vector information shall be calculated for CS7.

Table E-14. Numerical Values of Differential Chromaticities at CS4 for the Second Example LED

	$\delta u'$	$\delta v'$
CS4	0.003611	0.003739
CS7	N/A	N/A

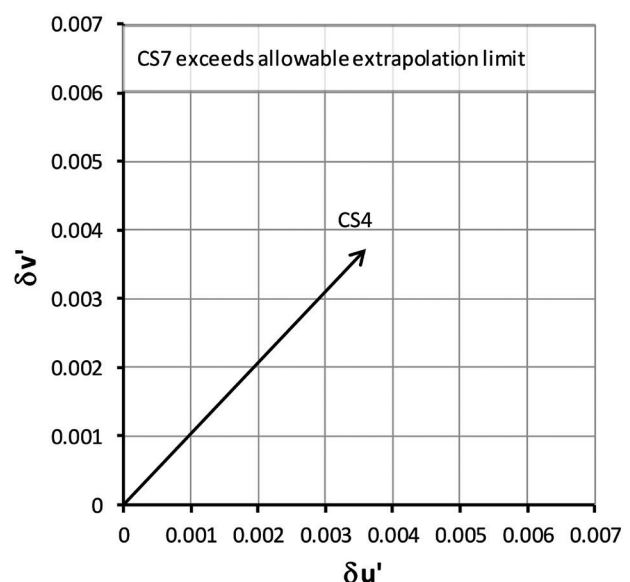


Figure E-4. Vector graphic for first example LED.

ADDITIONAL READING

Bergman R. Summary of NVLAP Round 15 Proficiency Test. CORM 2018 Annual Conference and Business Meeting, Gaithersburg, MD, July 29 – August 1, 2018.

Davis JL et al. Chromaticity maintenance in LED devices. In: van Driel W, Fan X, and Zhang GQ, editors. Solid State Lighting Reliability: Components to Systems, Vol. 2. New York: Springer; 2017.

Davis JL, Young J, Royer M. CALiPER Report 20.5: Chromaticity Shift Modes of LED PAR38 Lamps Operated in Steady-State Conditions. Pacific Northwest National Laboratory; 2016.

Mehr YM, van Driel WD, Zhang GQ. Progress in understanding color maintenance in solid-state lighting systems. Engineering. 2015;1(2):170-8.

Royer MP, Tuttle R, Rosenfeld S, Miller NJ, Color Maintenance in Laboratory and Field Applications, Pacific Northwest National Laboratory, 2013.

Wagner M, Herzog HG, Khanh TQ. Lifetime calculation of white HP-LEDs from 16,000 hours aging data. In: Luger Res EU. 2017;59:34-8.

Process for Change to an ANSI/IES Standard Under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard Under Continuous Maintenance

NOTE:

Use a separate form for each comment. Submit to the Director of Standards, IES, 85 Broad St, FL 17, New York NY 10004.
Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
Telephone: _____
Fax: _____
E-mail: _____

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



Lighting Science Standards

Fundamentals, Metrics and Calculations

Lighting Practice Standards

Design, Engineering, and Specifications

Lighting Applications Standards

Design Criteria and Illumination Recommendations

Lighting Measurements and Testing Procedure Standards

Industry Standardization

Roadway and Parking Facility Lighting Standards

Criteria and Illumination Recommendations

Order# ANSI/IES TM-35-19(R25)

ISBN# 978-0-87995-538-0

www.ies.org